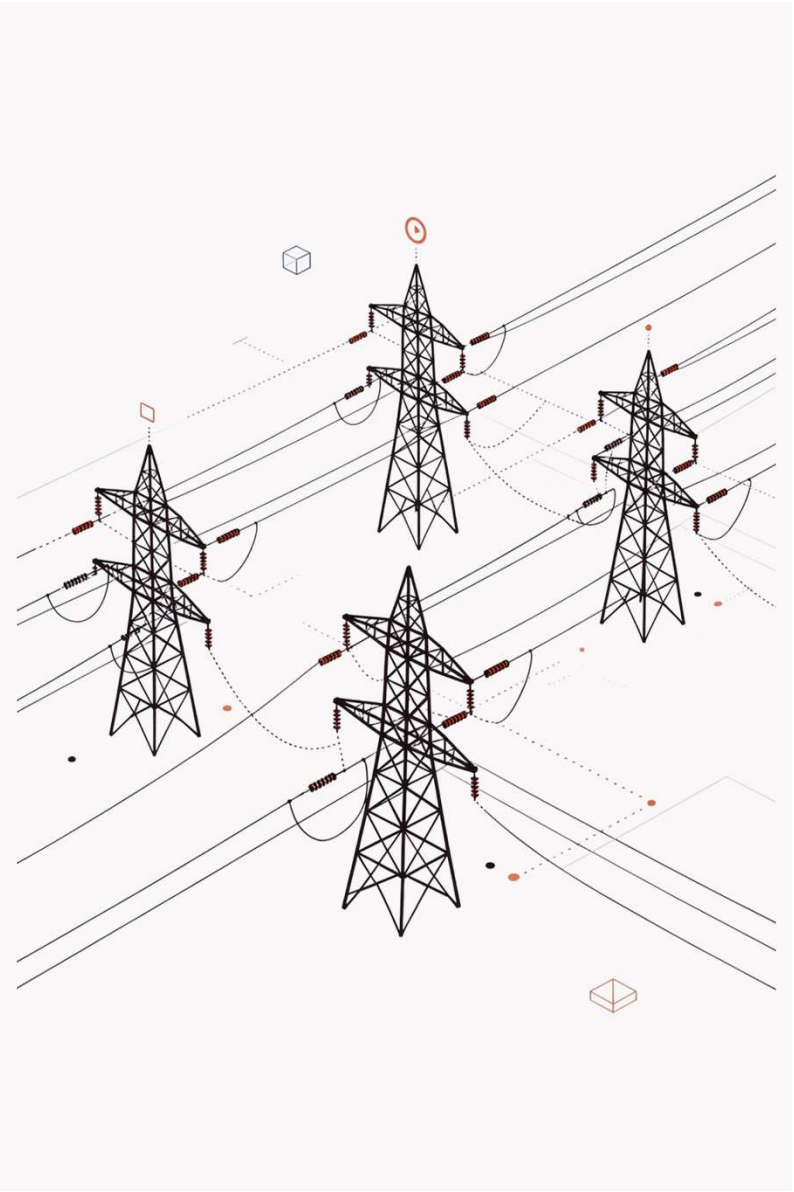
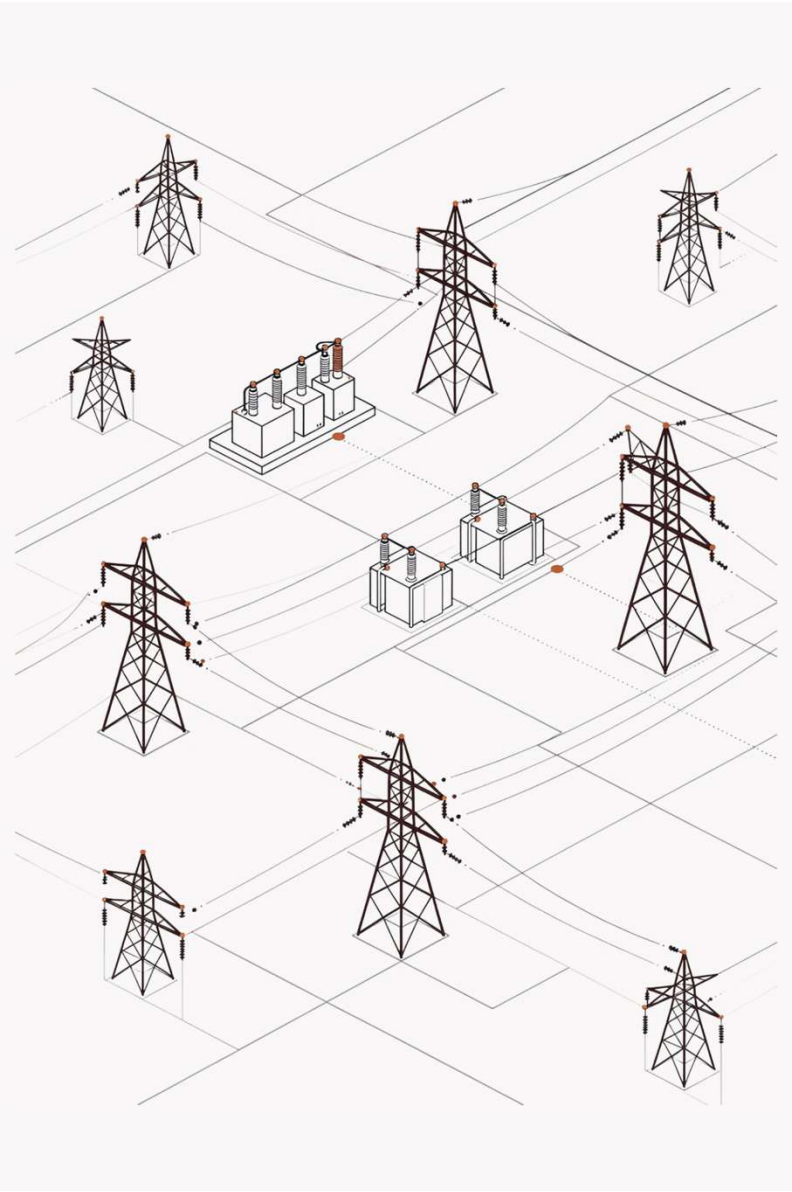


Power Transmission System – Electrical Model

Line Parameters: **Resistance, Inductance & Capacitance** Derivation for Three-Phase Double Circuit Lines

ELECTRICAL ENGINEERING | POWER SYSTEMS | LINE PARAMETERS



Why Line Parameters Matter

Distributed R, L, C — Not Ideal Wires

Transmission lines exhibit distributed resistance, inductance, and capacitance that directly affect voltage regulation, power flow, and system stability.

Foundation of Power System Analysis

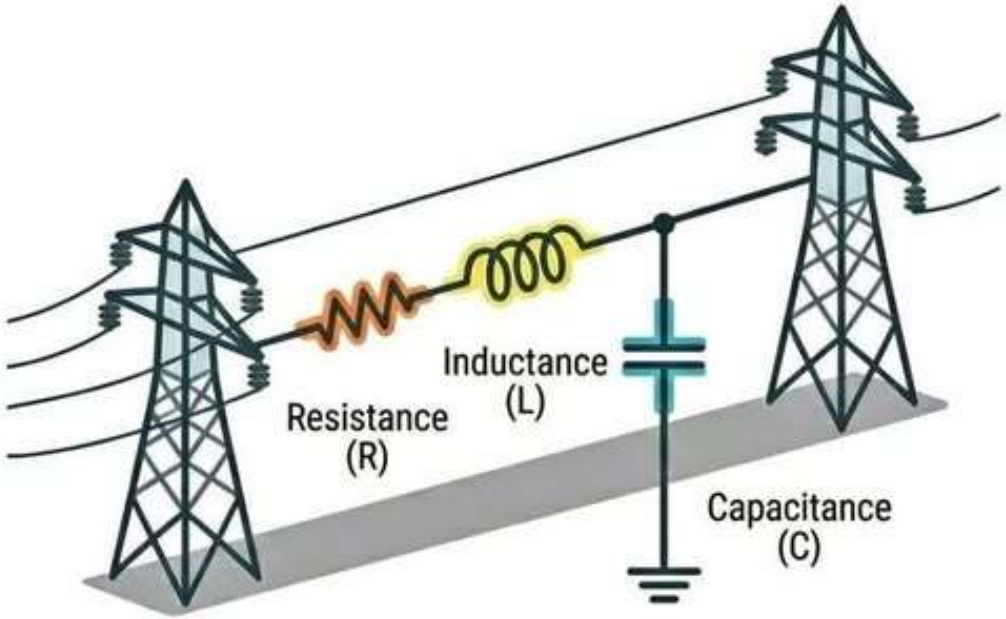
Accurate parameter derivation is essential for load flow studies, fault analysis, and system protection design — errors here propagate through every downstream calculation.

Double Circuit Lines — The Grid Backbone

Three-phase double circuit lines are the backbone of high-voltage grids: compact, efficient, and magnetically coupled between the two circuits sharing the same tower structure.

The Triple-Parameter Model

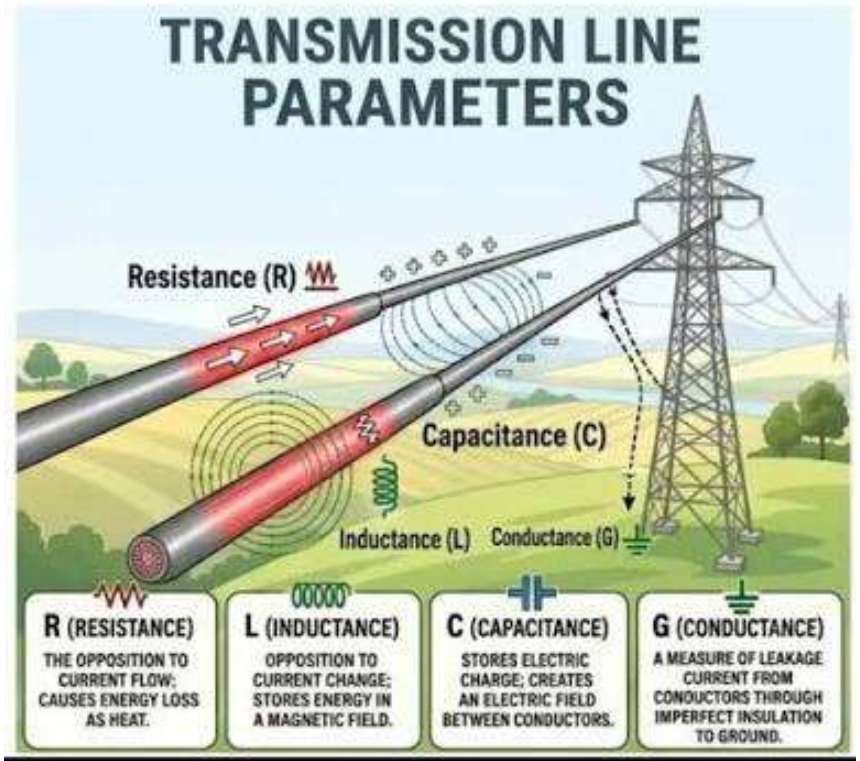
Lines are modeled with series R & L and shunt C.

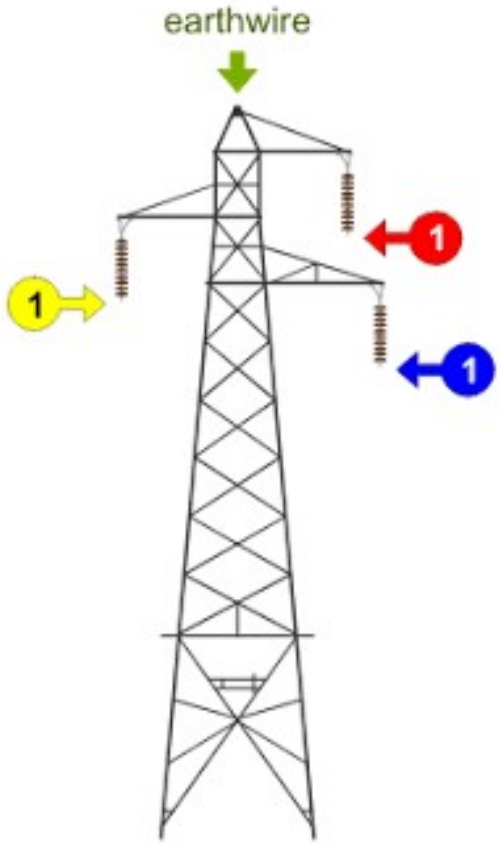


KEY MODEL PROPERTIES:

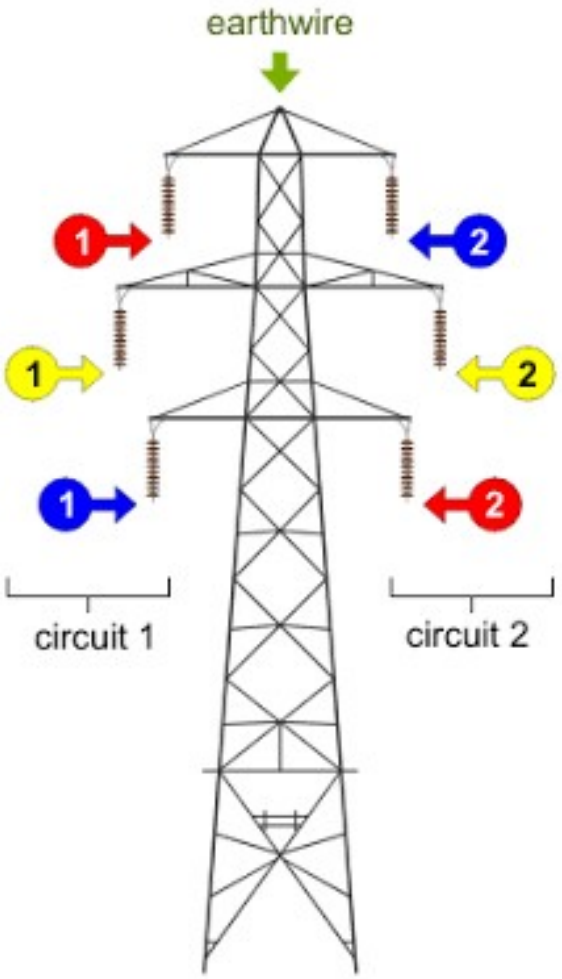
- Resistance (R): Represents line power losses (heat).
- Inductance (L): Models magnetic field and series impedance.
- Capacitance (C): Accounts for electric field and line charging.

TRANSMISSION LINE PARAMETERS - A PRACTICAL OVERVIEW





SEE PL1a S2 (single circuit)

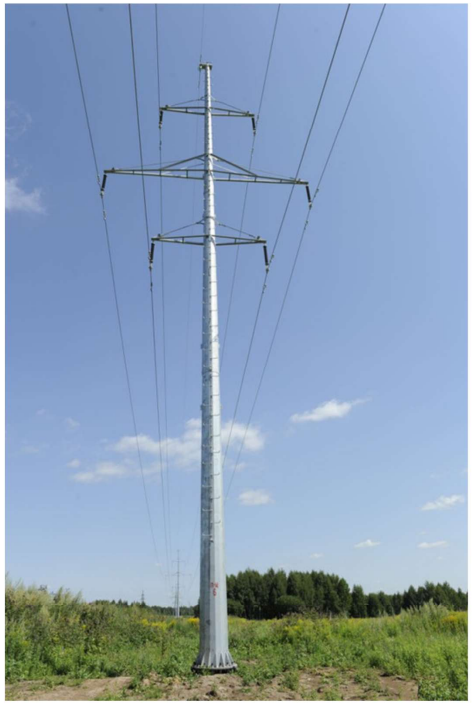


SEE PL1a D2 (double circuit)

Overhead line

- Single-circuit overhead lines
- Double-circuit overhead lines
- Multi-circuit overhead lines



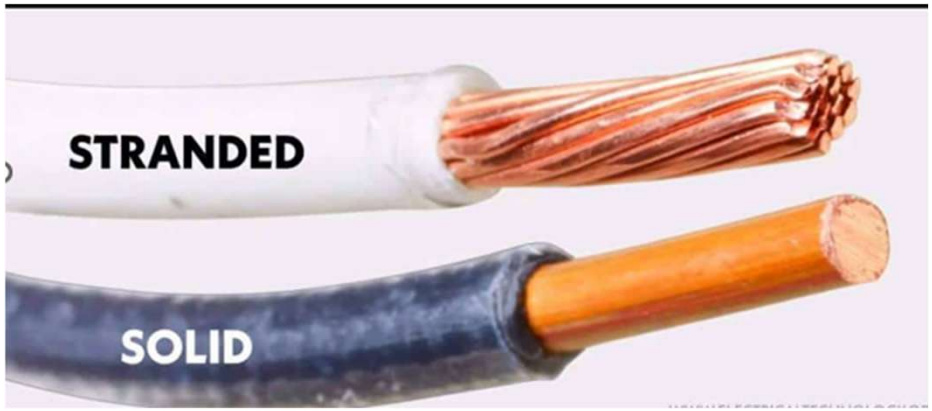


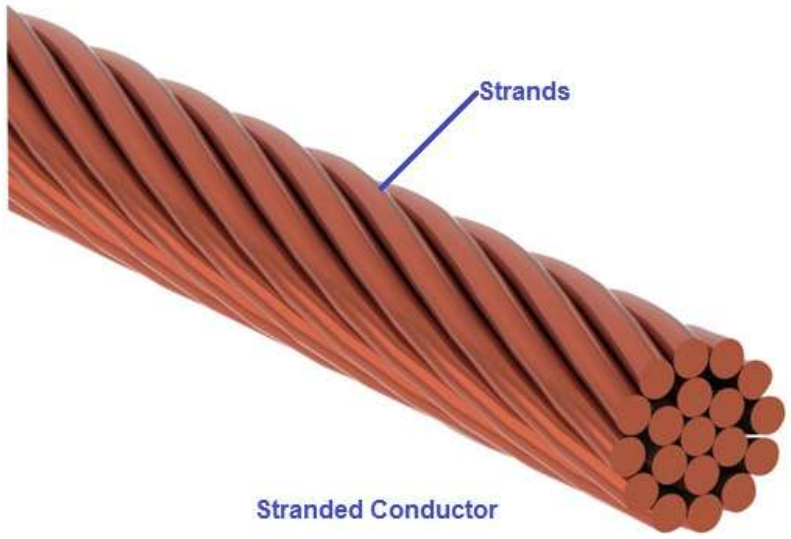


Stranded Conductors

- Stranded conductors consist of multiple thin wires helically twisted together to form a single, flexible cable.

- Widely used in overhead power systems, they offer
  - superior mechanical strength,
  - resistance to fatigue from wind-induced vibrations,
  - and ease of transportation over a solid wire.





Composite Conductors

- Composite conductors combine two or more distinct materials
- to merge the mechanical strength of one with the high electrical conductivity of another.
- They are widely used in overhead power transmission and distribution to drastically reduce line sag, increase capacity, and minimize power losses.

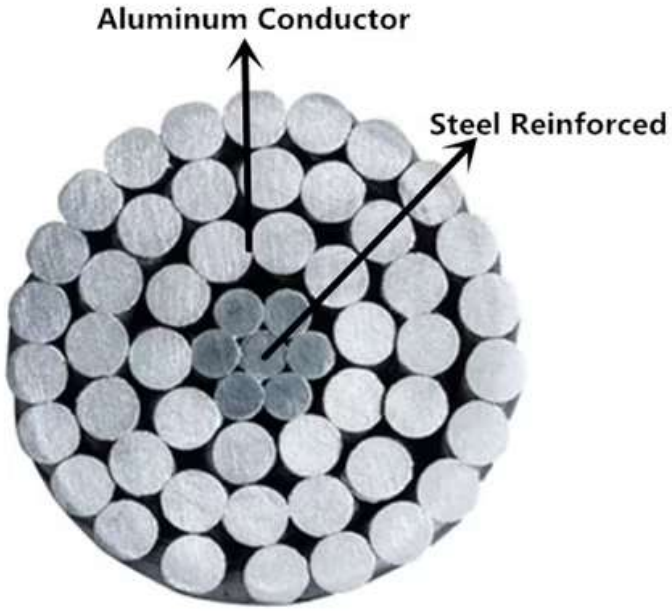


ACSR Conductors

- An ACSR (Aluminum Conductor Steel Reinforced) cable is an overhead power transmission and distribution wire.

- It features a central, high-strength galvanized steel core surrounded by strands of high-purity aluminum.

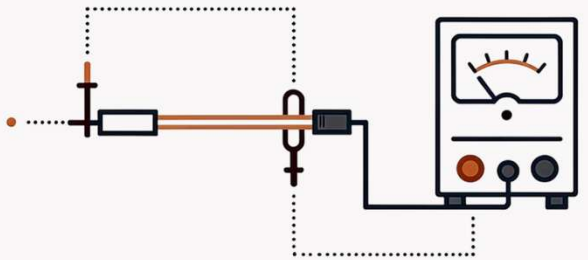
- The steel core provides mechanical tension, while the aluminum handles the electrical conductivity



CHAPTER 1

Resistance

DC resistance, skin effect, AC correction, and equivalent resistance for double circuit configurations.



DC Resistance of a Conductor

Fundamental Formula

The DC resistance of a conductor of length L and cross-sectional area A is:

Rdc = ρL / A (Ω)

where **ρ** is the resistivity in Ω·m. For copper ρ ≈ 1.72×10^{-8} Ω·m; for aluminum ρ ≈ 2.83×10^{-8} Ω·m.

Stranding Correction

For stranded conductors (e.g., ACSR), the effective area is slightly less than nominal due to the helical lay of strands — approximately a **~2% correction factor** applied to A.

Temperature Correction

If ρ_1 and ρ_2 are the values of resistivity corresponding to two different temperatures t_1 and t_2, then

$$\rho_2 = \rho_1[1 + \alpha\ (t_2 - t_1)]$$

where **α** is the temperature coefficient of resistance. For aluminum, α ≈ 0.00403 /°C. Operating temperature is typically taken as 75°C for ampacity calculations.

$$\alpha_1 = \frac{\alpha_0}{1 + \alpha_0 t_1}$$

where α_0 is the temperature coefficient of resistance at 0°C.

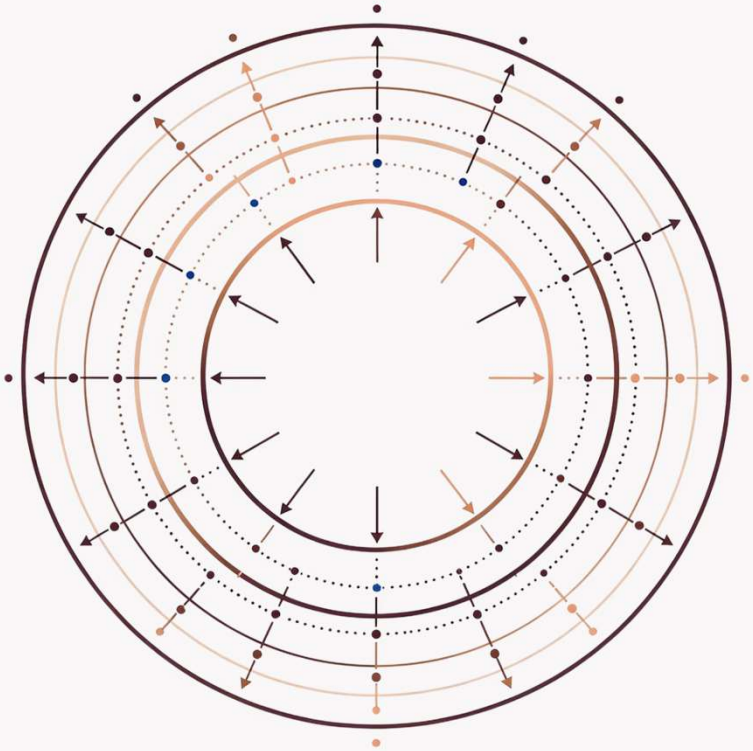
Transmission Line Resistance – Significance

- Transmission line resistance causes power loss in the form of heat (I^2R loss).
- It reduces transmission efficiency by dissipating a part of the transmitted power.
- Line resistance causes voltage drop along the transmission line, affecting voltage regulation.
- It limits the current-carrying capacity of the conductor due to temperature rise.
- Resistance increases with conductor temperature and influences the thermal rating of the line.
- Minimizing resistance through proper conductor selection helps reduce losses and improve the overall performance of the transmission system.

AC Resistance — Skin Effect

Skin Effect Mechanism

At power frequency (50/60 Hz), alternating current **crowds toward the conductor surface**, reducing the effective cross-sectional area and increasing resistance above the DC value.



Rac vs Rdc

AC resistance is always **greater than DC resistance**. The ratio Rac/Rdc increases with conductor diameter and frequency.
Bundled conductors reduce skin effect and corona losses while increasing effective GMR.



Cross-sectional area of a round conductor available for conducting DC current

"DC resistance"

Cross-sectional area of the same conductor available for conducting low-frequency AC

"AC resistance"

Cross-sectional area of the same conductor available for conducting high-frequency AC

"AC resistance"

Resistance of a Double Circuit Line

1

Two Circuits in Parallel

When both circuits carry the same phase current (identical conductors), the equivalent phase resistance is simply:

> **Req = Rphase / 2** (Ω/km)

Each phase conductor resistance Rphase is taken from manufacturer data tables at operating temperature.

2

Bundled Sub-Conductors per Phase

For bundled conductor configurations (n sub-conductors per phase):

> **Rbundle = Rconductor / n**

Combining with the parallel circuit effect:

> **Req = Rconductor / (2n)** (Ω/km)

3

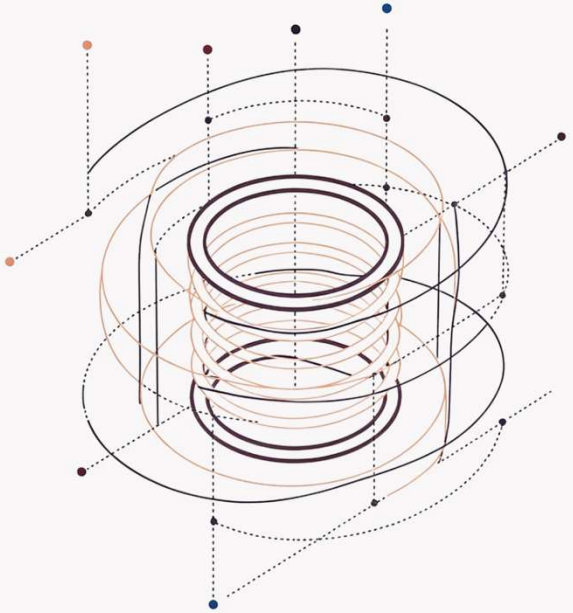
Practical Values

Typical 765 kV double circuit lines use **n = 4** sub-conductors per phase bundle (quad bundle). ACSR conductor resistance at 75°C is typically **0.05–0.10 Ω/km** per sub-conductor.

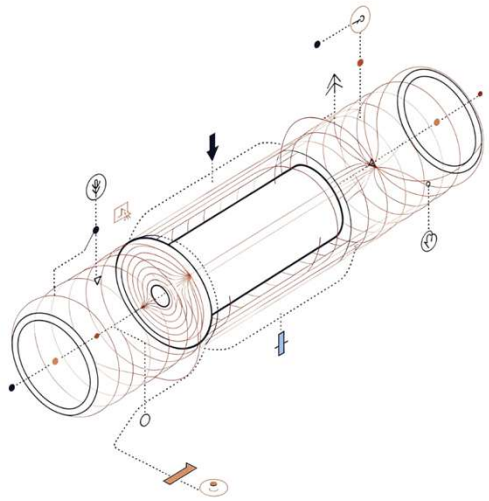
CHAPTER 2

Inductance

Flux linkage fundamentals, GMR and GMD concepts, and full derivation for three-phase double circuit lines.



Flux Linkage and Inductance — Fundamentals



Definition of Inductance

Inductance arises from the magnetic flux linking a current-carrying conductor:

> $\mathbf{L = \lambda / I}$ (H/m)

Total Flux Linkage of Conductor i

Summing contributions from all conductors in the system:

> $\mathbf{\lambda i = (\mu_0/2\pi)\ \Sigma\ Ij\ ln(1/Dij)}$

where Dij is the distance between conductors i and j (or the GMR when i = j).

Internal Flux Linkage

For a solid round conductor, the internal flux linkage contributes an additional $\mathbf{\mu_0/8\pi}$ **H/m**, which is accounted for by replacing radius r with:

> $\mathbf{GMR = r \cdot e{-}1/4 = 0.7788r}$

Transmission Line Inductance – Significance

- Transmission line inductance stores energy in the magnetic field surrounding the conductors.

- It produces inductive reactance, which causes voltage drop along the transmission line.

- Line inductance absorbs reactive power and affects the voltage regulation of the system.

- It limits the maximum power that can be transmitted through the line.

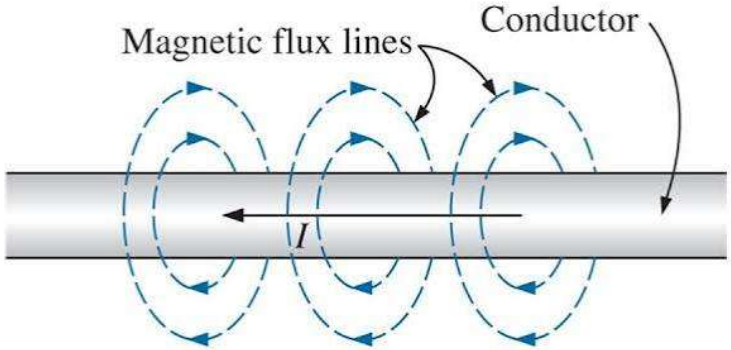
- It influences short-circuit current, protection requirements, and circuit-breaker ratings.

- Inductance affects surge impedance, system stability, and the overall design of EHV transmission lines.

Flux Linkage

• When current flows through a transmission line conductor, it produces a magnetic field around the conductor.
• The magnetic field consists of magnetic flux lines surrounding the conductor.
• These magnetic flux lines link the conductor carrying the current.
• The total magnetic flux linking a conductor is called flux linkage.

Magnetic flux lines around a current-carrying conductor.

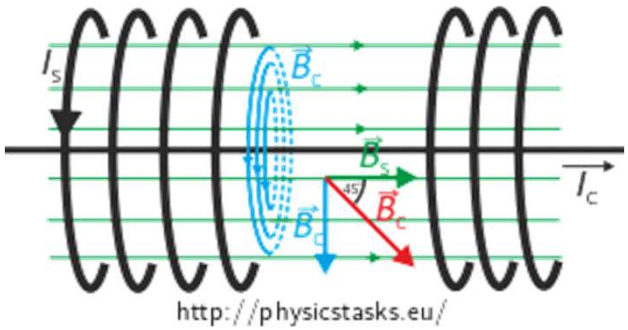


Definition

Flux linkage is the total magnetic flux that links a conductor due to the current flowing through it.

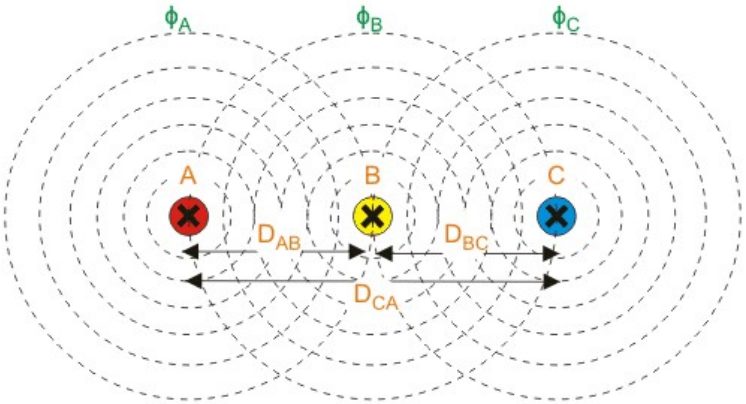
Unit: Weber-turn (Wb-turn)

Types of Flux Linkage



Internal Flux Linkage

- Produced by the magnetic flux inside the conductor.
- Exists because the conductor has a finite radius.
- Depends on the current distribution within the conductor.
- Contributes only a small portion of the total inductance.



External Flux Linkage

- Produced by the magnetic flux outside the conductor.
- Extends from the conductor surface to the return path.
- Depends on conductor spacing and arrangement.
- Contributes the major portion of the transmission line inductance.

Flux Linkage and Inductance

Flux Linkage Equation

$$\lambda = N\Phi$$

λ = Flux linkage (Wb-turn)
N = Number of turns
Φ = Magnetic flux (Wb)

For a transmission line conductor, N = 1

Therefore, $\lambda = \Phi$

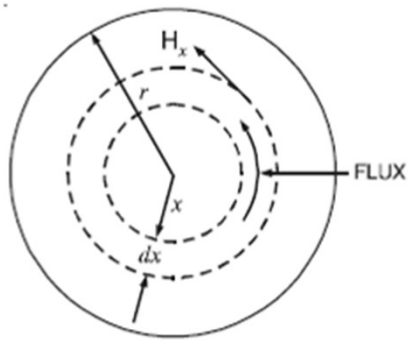
Relation with Inductance

$$L = \lambda / I$$

L = Inductance (H)
I = Current (A)

Methodology for Deriving Internal Flux Linkage

- Assume the conductor carries a uniformly distributed current.
- Consider a thin cylindrical shell of radius x and thickness dx inside the conductor.
- Calculate the current enclosed within radius x.
- Apply Ampere's Circuital Law to determine the magnetic field intensity (H).
- Determine the magnetic flux density using: B = μH
- Calculate the elemental flux (dΦ) through the shell.
- Since only the enclosed current links this flux, determine the elemer



- Integrate dψ from the centre (x = 0) to the conductor surface (x = r) to obtain the total internal flux linkage.

Flux Linkages of a Conductor Due to Internal Flux. Consider a long straight cylindrical conductor of radius r metres and carrying a current of I amperes, as shown in Fig. 5.3.

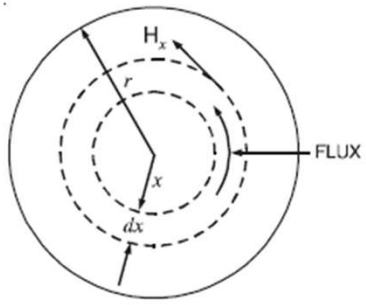


Fig. 5.3

In overhead lines it may be assumed without appreciable error that the current is uniformly distributed.

The current inside a line of force of radius x,

$$I_x = \frac{I}{\pi r^2} \times \pi x^2 = \frac{Ix^2}{r^2}$$

Field strength inside the conductor at a distance x from the centre,

$$H_x = \frac{\text{Current}}{2\pi \times \text{distance}} = \frac{Ix^2}{r^2} \times \frac{1}{2\pi x} = \frac{Ix}{2\pi r^2} \text{ AT/m}$$

Flux density,

$B_x = \mu_0 \mu H_x$ Wb/m^2 or T where $\mu_0 = 4\pi \times 10^{-7}$

and μ is the relative permeability of the medium and for non-magnetic materials $\mu = 1$.

$$\therefore B_x = \frac{\mu_0 Ix}{2\pi r^2} \text{ Wb/m}^2$$

$$d\phi = B_x \times 1 \times dx = \frac{\mu_0 x}{2\pi r^2} \cdot I \cdot dx \text{ webers}$$

But this flux links with only the current lying within the circle of radius x, *i.e.*, with I_x or $\frac{x^2}{r^2} I$

$\therefore$ Linkages of the shell,

$$d\psi = \frac{\mu_0 x}{2\pi r^2} \cdot dx \times \frac{x^2}{r^2} \times I = \frac{\mu_0 I x^3}{2\pi r^4} dx \text{ weber-turns}$$

Total flux linkages from centre of the conductor and up to the surface of the conductor,

$$\psi_{int} = \int_0^r \frac{\mu_0 I x^3}{2\pi r^4} dx \text{ Wb-turns/m}$$

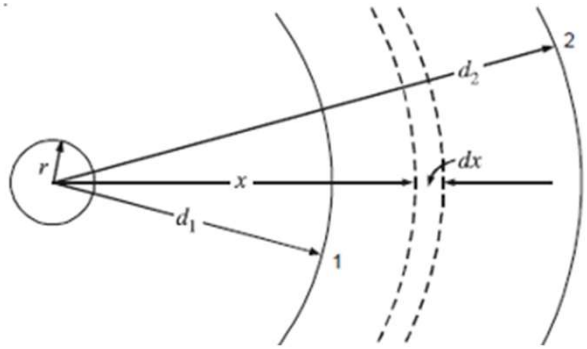
$$= \frac{\mu_0 I}{8\pi} \text{ Wb-turns/m}$$

$$= \frac{4\pi \times 10^{-7} \times I}{8\pi} \text{ Wb-turns/m}$$

$$= \frac{1}{2} I \times 10^{-7} \text{ Wb-turns/m}$$

Methodology for Deriving External Flux Linkage

- Consider a thin cylindrical shell of radius x and thickness dx outside the conductor.
- Apply Ampere's Circuital Law to determine the magnetic field intensity (H).
- Calculate the magnetic flux density using: B = μH
- Determine the elemental flux (dΦ) through the shell.



- Since external flux links the entire conductor current, the elemental flux linkage is equal to the elemental flux.
- Integrate dψ between the specified radial distances (d1 to d2) to obtain the total external flux linkage.

Flux Linkages of a Conductor Due to External Flux. Consider two points 1 and 2 distant d_1 and d_2 from the centre of the conductor. Since the flux paths are concentric circles around the conductor, whole of the flux between points 1 and 2 lies within the concentric cylindrical surfaces passing through these points 1 and 2.

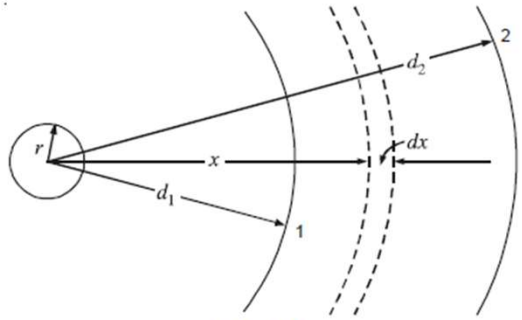


Fig. 5.4

The field strength at any distance x from the centre of the conductor ($x > r$),

$$H_x = \frac{I}{2\pi x} \text{ AT/m}$$

Flux density,

$$B_x = \mu_0 H_x = \frac{\mu_0 I}{2\pi x} \text{ Wb/m}^2$$

So the flux through a cylindrical shell of radial thickness dx and axial length one metre

$$d\phi = \frac{\mu_0 I}{2\pi x}\, dx \text{ webers per metre}$$

Now flux linkages per metre is equal to $d\phi$ since flux external to conductor links all the current in the conductor once and only once

$$\therefore \quad d\psi = \frac{\mu_0 I}{2\pi x}\, dx \text{ Wb-turns per metre}$$

Total flux linkages between points 1 and 2

$$\psi_{ext} = \int_{d_1}^{d_2} \frac{\mu_0 I}{2\pi x}\, dx = \frac{4\pi \times 10^{-7}}{2\pi} I \left[\log_e x\right]_{d_1}^{d_2}$$

$$= 2 \times 10^{-7} I \log_e \frac{d_2}{d_1} \text{ Wb-turns/m}$$

Total Flux Linkage of a Conductor

The total flux linkage is the sum of the internal and external flux linkages.

$$\psi = \psi_{int} + \psi_{ext}$$

Substituting the expressions,

$$\psi = (\mu_0 I/8\pi) + (\mu_0 I/2\pi) \ln(D/r)$$

Taking $(\mu_0 I/2\pi)$ as common,

$$\psi = (\mu_0 I/2\pi) [\ln(D/r) + 1/4]$$

$$\frac{1}{4} = \ln\left(e^{1/4}\right)$$

Since

$$e^{1/4} \approx 1.2840$$

we have

$$\ln\left(\frac{D}{r}\right) + \frac{1}{4} = \ln\left(\frac{D}{r}\right) + \ln\left(e^{1/4}\right)$$
$$= \ln\left(\frac{De^{1/4}}{r}\right)$$

Now define

$$r' = \frac{r}{e^{1/4}}$$

Since

$$\frac{1}{e^{1/4}} = e^{-1/4} = 0.7788$$

we obtain

$$r' = 0.7788r$$

Simplification Using Equivalent Radius

Since,

$\ln(D/r) + 1/4 = \ln(D/r')$

$r' = 0.7788\ r$

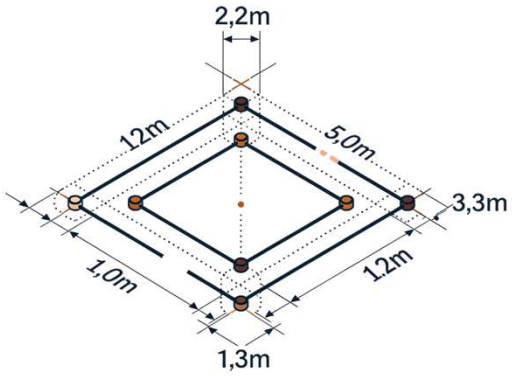
Therefore,

$$\psi = (\mu_0 I/2\pi) \ln(D/r')$$

Where, r' = Geometric Mean Radius (GMR) or equivalent radius.

Note:
The use of r' incorporates the effect of internal flux linkage, allowing the total flux linkage to be expressed in a compact form.

GMR and GMD — Key Concepts



Geometric Mean Radius (GMR)

The GMR is the **effective radius** that accounts for internal flux linkage. For a solid round conductor:

> **$GMR = r \cdot e^{-1/4} \approx 0.7788r$**

For ACSR and stranded conductors, GMR is tabulated by manufacturers — it is always less than the physical radius r.

Geometric Mean Distance (GMD)

The GMD (or Deq) replaces the actual unequal inter-conductor spacings Dab, Dbc, Dca after transposition:

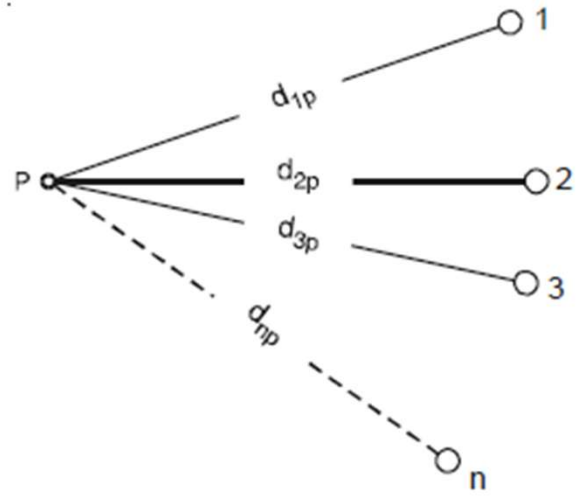
> **$D_{eq} = (D_{ab} \cdot D_{bc} \cdot D_{ca})^{1/3}$**

Physical Significance

GMD is the equivalent uniform spacing that produces the same average flux linkage as the actual asymmetric arrangement — valid only for a **fully transposed** line.

Flux Linkages of One Conductor in a Group of Conductors

Consider a group of parallel conductors 1, 2, 3..... *n* carrying currents I_1, I_2, I_3 ..... I_n respectively, as illustrated in Fig.
Let it be assumed that the sum of the currents in various conductors is zero *i.e.*, $I_1 + I_2 + I_3 + ... I_n = 0$.



Principle Behind the Derivation

• Every current-carrying conductor produces a magnetic field around it.

• The magnetic field produced by each conductor links not only itself but also the neighboring conductors.

• Hence, the total flux linkage of a conductor consists of:
  – Self flux linkage due to its own current.
  – Mutual flux linkages due to currents in all other conductors.

• Using the principle of superposition, the total flux linkage is obtained by summing the individual flux linkages produced by all conductors.

• The derivation assumes that the algebraic sum of currents in all conductors is zero ($\Sigma I = 0$), which simplifies the final expression.

The flux linkages of conductor 1 due to its own current I_1 (internal and external), up to point P

$$\psi_{1p1} = 2 \times 10^{-7} \, I_1 \log_e \frac{d_{1p}}{r_1'} \text{ Wb-turns/metre}$$

The flux linkages of conductor 1 due to current in conductor 2

$$\psi_{1p2} = 2 \times 10^{-7} \, I_2 \log_e \frac{d_{2p}}{d_{12}}$$

Flux due to conductor 2 that lies between conductors 2 and 1 does not link conductor 1 and therefore the distances involved are d_{2p} and d_{12}.

Thus, the expression for flux linkages of conductor 1 due to currents in all conductors can be written as

$$\psi_{1p} = 2 \times 10^{-7} \left[I_1 \log_e \frac{d_{1p}}{r_1'} + I_2 \log_e \frac{d_{2p}}{d_{12}} + I_3 \log_e \frac{d_{3p}}{d_{13}} + \ldots I_n \log_e \frac{d_{np}}{d_{1n}} \right]$$

Wb-turns/m ...(5.10)

The above equation may be written as

$$\psi_{1p} = 2 \times 10^{-7} \left[I_1 \log_e \frac{1}{r_1'} + I_2 \log_e \frac{1}{d_{12}} + I_3 \log_e \frac{1}{d_{13}} + \ldots I_n \log_e \frac{1}{d_{1n}} \right]$$
$$+ 2 \times 10^{-7} \left[I_1 \log_e d_{1p} + I_2 \log_e d_{2p} + I_3 \log_e d_{3p} \right.$$
$$\left. + \ldots I_n \log_e d_{np} \right] \text{ Wb-turns/m} \quad \ldots(5.11)$$

The above equation may be written as

$$\psi_{1p} = 2 \times 10^{-7} \left[I_1 \log_e \frac{1}{r_1'} + I_2 \log_e \frac{1}{d_{12}} + I_3 \log_e \frac{1}{d_{13}} + \ldots I_n \log_e \frac{1}{d_{1n}} \right]$$
$$+ 2 \times 10^{-7} \left[I_1 \log_e d_{1p} + I_2 \log_e d_{2p} + I_3 \log_e d_{3p} \right.$$
$$\left. + \ldots I_n \log_e d_{np} \right] \text{ Wb-turns/m} \quad \ldots(5.11)$$

To account for the total flux linkages to conductor 1, the point P must approach infinity and in this condition

$$d_{1p} \simeq d_{2p} \simeq d_{3p} \simeq d_{4p} \ldots d_{np} \simeq d \text{ (say)}$$

Then, $\lim_{d \to \infty} (I_1 + I_2 + I_3 + \ldots + I_n) \log_e d = 0$

$$\because \; I_1 + I_2 + I_3 + \ldots I_n = 0$$

This simplifies the Eq. (5.11) and the equation for the flux linkages to conductor 1 becomes

$$\psi_1 = 2 \times 10^{-7} \left[I_1 \log_e \frac{1}{r_1'} + I_2 \log_e \frac{1}{d_{12}} + I_3 \log_e \frac{1}{d_{13}} + \ldots I_n \log_e \frac{1}{d_{1n}} \right]$$

Wb-turns/m ...(5.12)

Inductance of Composite Conductors

- A composite conductor consists of several strands connected electrically in parallel.

- Current divides among all the strands

- Each strand is linked by:
  - Flux due to its own current (self flux).
  - Flux due to currents in all the other strands (mutual flux).

- The total flux linkage of a strand is obtained by applying the principle of superposition.

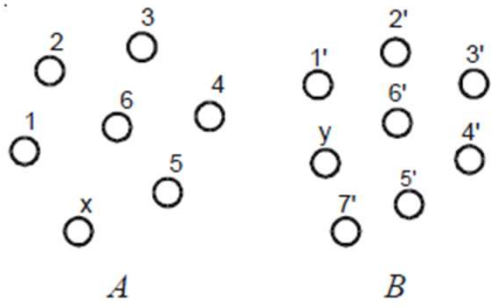
- The average flux linkage of all the strands is then used to determine the inductance of the composite conductor.

- This leads to the concepts of Self GMD (GMR) and Mutual GMD, which simplify the final inductance expression.

Inductance of Composite Conductor Lines—Self and Mutual GMDs

Consider a single phase line consisting of two parallel conductors A and B, conductor A consisting of x and conductor B of y strands, as illustrated in Fig. 5.7.

Let the conductors A and B carry currents I and –I respectively (since conductors of a 2-wire line carry the same current but in opposite directions).



Assuming uniform current density in both the conductors the current carried by each strand of conductor A will be $\frac{I}{x}$ while that carried by each strand of conductor B will be $\frac{-I}{y}$

Using Eq. (5.12) the flux linkages of strand 1 in conductor A can be written as

$$\psi_1 = 2 \times 10^{-7} \frac{I}{x}\left[\log_e \frac{1}{r_1'} + \log_e \frac{1}{d_{12}} + \log_e \frac{1}{d_{13}} + \ldots + \log_e \frac{1}{d_{1x}}\right]$$

$$- 2 \times 10^{-7} \frac{I}{y}\left[\log_e \frac{1}{d11'} + \log_e \frac{1}{d12'} + \log_e \frac{1}{d13'} + \ldots + \log_e \frac{1}{d_{1y}}\right]$$

$$\text{or } \psi_1 = 2 \times 10^{-7} \, I \log_e \frac{\sqrt[y]{(d_{11'} \; d_{12'} \; d_{13'} \ldots d_{1y})}}{\sqrt[x]{r_1' \; d_{12} \; d_{13} \ldots d_{1x}}} \text{ Wb-turns/m}$$

Inductance of strand 1 of conductor A

$$L_1 = \frac{\psi_1}{\frac{I}{x}} = 2 \times 10^{-7} \, x \log_e \frac{\sqrt[y]{(d_{11'} \; d_{12'} \; d_{13'} \; d_{14'} \ldots d_{1y})}}{\sqrt[x]{r' \; d_{12} \; d_{13} \; d_{14} \ldots d_{1x}}} \quad \ldots(5.13)$$

$$L_1 = 2 \times 10^{-7} x \log_e \frac{\sqrt[y]{d_{11'}\, d_{12'}\, d_{13'}\, d_{14'} \ldots . d_{1y}}}{\sqrt[x]{d_{11}\, d_{12}\, d_{13}\, d_{14} \ldots . d_{1x}}} \text{ H/m} \quad \ldots(5.14)$$

Similarly the expression for inductance for strand 2 can be written as

$$L_2 = 2 \times 10^{-7} x \log_e \frac{y\sqrt{d_{21'}\, d_{22'}\, d_{23'}\, d_{24'} \ldots . d_{2y}}}{x\sqrt{d_{21}\, d_{22}\, d_{23}\, d_{24} \ldots . d_{2x}}} \text{ H/m} \quad \ldots(5.15)$$

$$L_{av} = \frac{L_1 + L_2 + L_3 + \ldots . L_x}{x}$$

Since x such strands of conductor A are electrically in parallel,

Therefore inductance of conductor A,

$$L_A = \frac{L_{av}}{x}$$

$$= \frac{L_1 + L_2 + L_3 + \ldots L_x}{x^2}$$

$$= 2 \times 10^{-7} \log_e \left[\frac{\sqrt[xy]{\begin{matrix}(d_{11'}\, d_{12'}\, d_{13'}\, d_{14'} \ldots d_{1y})(d_{21'}\, d_{22'}\, d_{23'} \ldots d_{2y}) \\ \ldots (d_{x1'}\, d_{x2'}\, d_{x3'} \ldots d_{xy})\end{matrix}}}{x^2\sqrt{\begin{matrix}(d_{11}\, d_{12}\, d_{13}\, d_{14} \ldots d_{1x})(d_{21}\, d_{22}\, d_{23} \ldots d_{2x}) \\ (d_{x1}\, d_{x2}\, d_{x3} \ldots dxx)\end{matrix}}} \right] \quad \ldots(5.16)$$

$$\therefore \quad L_A = 2 \times 10^{-7} \log_e \frac{D_m}{D_{sA}} \text{ H/m}$$

Similarly $L_B = 2 \times 10^{-7} \log_e \frac{D_m}{D_{sB}}$ H/m

If conductors A and B are identical

$\therefore \quad D_{sA} = D_{sB} = D_s$ (say)

Loop inductance,

$$L = L_A + L_B$$

$$= 2 \times 10^{-7} \log_e \frac{D_m}{D_s} + 2 \times 10^{-7} \log_e \frac{D_m}{D_s}$$

$$= 4 \times 10^{-7} \log_e \frac{D_m}{D_s} \text{ H/m or } 0.4 \log_e \frac{D_m}{D_s} \text{ mH/km}.$$

INDUCTANCE OF THREE-PHASE OVERHEAD LINES

With Unsymmetrical Spacing. Consider a 3-ϕ line with conductors A, B, and C; each of radius r metres. Let the spacing between them be d_1, d_2 and d_3 and the current flowing through them be I_A, I_B and I_C respectively.

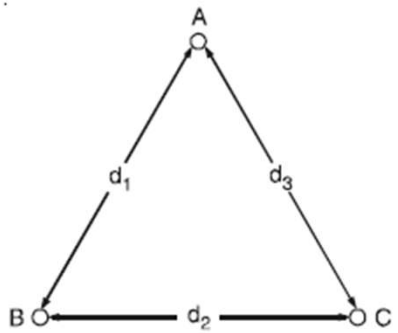


Fig. 5.11. 3-ϕ *Transmission Line With Unsymmetrical Spacing*



Principle Behind the Derivation

- Each phase conductor produces a magnetic field that links itself and the other two-phase conductors.

- The flux linkage of a phase consists of:
  - Self flux linkage due to its own current.
  - Mutual flux linkages due to the currents in the other two phases.

- Using the principle of superposition, the total flux linkage of each phase is obtained by summing the individual flux linkages.

- For a balanced three-phase system, the phase currents are equal in magnitude and displaced by 120°.

- Since the conductor spacing is unsymmetrical, the inductance of each phase is different.

- Transposition of the line is employed to equalize the average inductance of all three phases.

$$\psi_A = 2 \times 10^{-7} \left[I_A \log_e \frac{1}{r'} + I_B \log_e \frac{1}{d_1} + I_C \log_e \frac{1}{d_3} \right] \text{ Wb-turns/m}$$

Similarly,

$$\psi_B = 2 \times 10^{-7} \left[I_B \log_e \frac{1}{r'} + I_A \log_e \frac{1}{d_1} + I_C \log_e \frac{1}{d_2} \right] \text{ Wb-turns/m}$$

and

$$\psi_C = 2 \times 10^{-7} \left[I_C \log_e \frac{1}{r'} + I_A \log_e \frac{1}{d_3} + I_B \log_e \frac{1}{d_2} \right] \text{ Wb-turns/m}$$

If the system is balanced,

$I_A = I_B = I_C = I$ (say) in magnitude

Taking I_A as a reference phasor, the currents are represented, in symbolic form as

$\mathbf{I}_A = I$; $\mathbf{I}_B = I(-0.5 - j0.866)$ and $\mathbf{I}_C = I(-0.5 + j0.866)$

Substituting these values of I_B and I_C in the expression for ψ_A we get

$$\psi_A = 2 \times 10^{-7} \left[I \log_e \frac{1}{r'} + I(-0.5 - j0.866) \log_e \frac{1}{d_1} + I(-0.5 + j0.866) \log_e \frac{1}{d_3} \right]$$

$$= 2 \times 10^{-7} I \left[\log_e \frac{1}{r'} + \log_e \sqrt{d_1 d_3} + j\sqrt{3} \log_e \sqrt{\frac{d_1}{d_3}} \right]$$

and $L_A = \dfrac{\psi_A}{I_A}$

$$= 2 \times 10^{-7} \left[\log_e \frac{1}{r'} + \log_e \sqrt{d_1 d_3} + j\sqrt{3} \log_e \sqrt{\frac{d_1}{d_3}} \right] \text{ H/m} \quad \ldots(5.18)$$

Similarly,

$$L_B = 2 \times 10^{-7} \left[\log_e \frac{1}{r'} + \log_e \sqrt{d_1 d_2} + j\sqrt{3} \log_e \sqrt{\frac{d_2}{d_1}} \right] \text{ H/m} \quad \ldots(5.19)$$

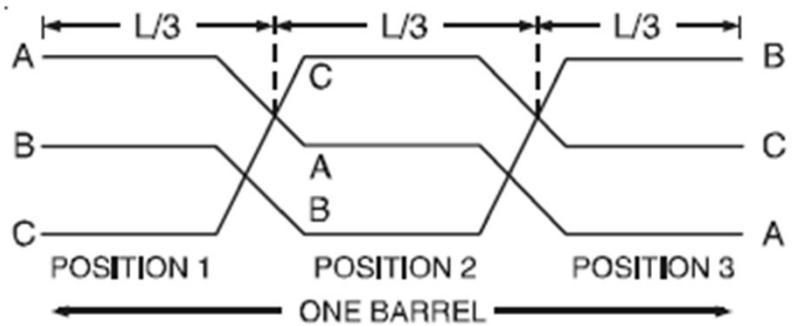
and $L_C = 2 \times 10^{-7} \left[\log_e \frac{1}{r'} + \log_e \sqrt{d_2 d_3} + j\sqrt{3} \log_e \sqrt{\frac{d_3}{d_2}} \right]$ H/m

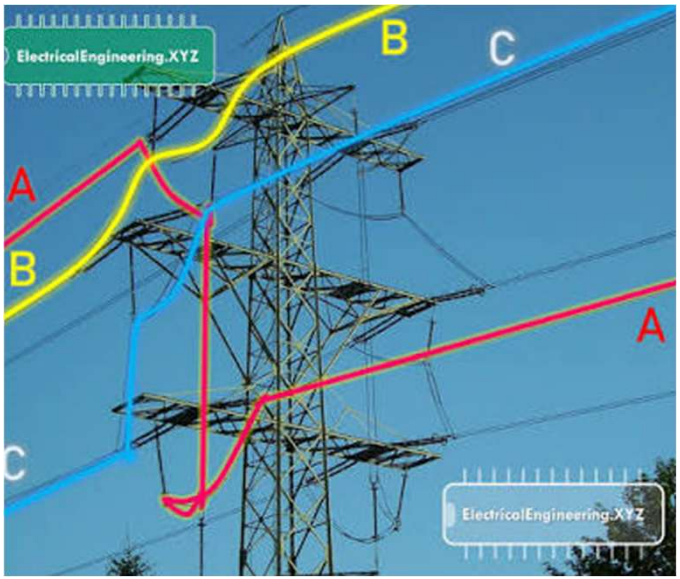
If conductors are equispaced:

$$\mathbf{L} = 2 \times 10^{-7} \log_e \frac{\sqrt[3]{d \times d \times d}}{r'} \text{ H/m}$$

$$= 2 \times 10^{-7} \log_e \frac{d}{r'} \text{ H/m}$$

Transposition of a 3-Phase Line





$$\mathbf{L} = \frac{L_A + L_B + L_C}{3}$$

$$= 2 \times 10^{-7}\left[\log_e \frac{1}{r'} + \frac{1}{3}\left(\log_e \sqrt{d_1 d_3} + \log_e \sqrt{d_1 d_2} + \log_e \sqrt{d_2 d_3}\right)\right.$$

$$\left. + \frac{j\sqrt{3}}{3}\left(\log_e \sqrt{\frac{d_1}{d_3}} + \log_e \sqrt{\frac{d_2}{d_1}} + \log_e \sqrt{\frac{d_3}{d_2}}\right)\right] \text{ H/m}$$

$$= 2 \times 10^{-7}\left[\log_e \frac{1}{r'} + \log_e \sqrt[3]{d_1\, d_2\, d_3} + 0\right]$$

$$= 2 \times 10^{-7} \log_e \frac{\sqrt[3]{d_1\, d_2\, d_3}}{r'} \text{ H/m} \qquad \ldots(5.21)$$

3-Phase Double-Circuit Line

- Two independent 3-phase circuits are carried on the same tower.
- Increases power transfer without a new transmission corridor.
- Improves reliability and continuity of supply.
- Reduces right-of-way, tower and construction costs.
- Mutual coupling improves line electrical characteristics.

Electrical Advantages

- Lower inductance.
- Higher capacitance.
- Lower surge impedance.
- Higher SIL and power transfer capability.



Inductance of a 3-Phase Double Circuit Line With Symmetrical Spacing

Consider a 3-phase double circuit connected in parallel-conductors A, B, C forming one circuit and conductors A', B', C' forming the other one, as illustrated in Fig. 5.19 (conductors symmetrically spaced).

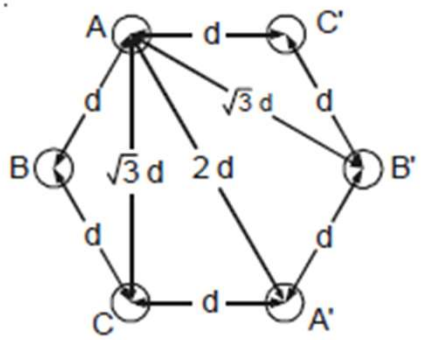
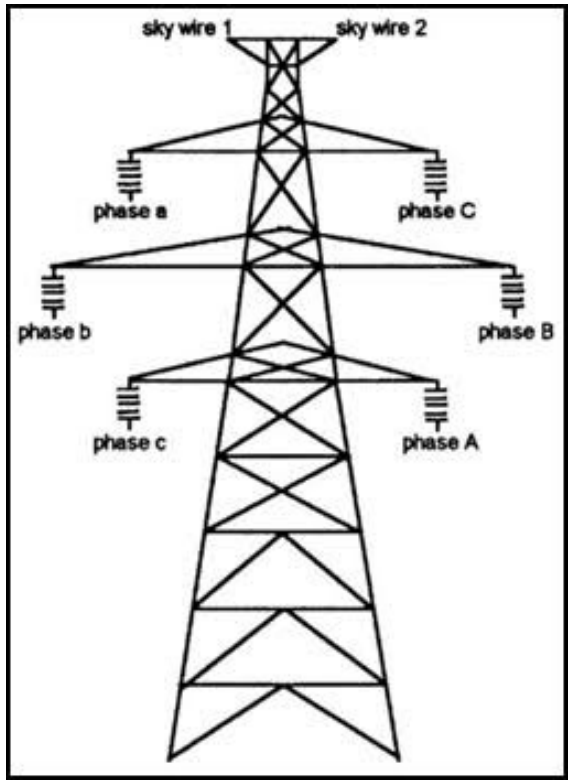


Fig. 5.19



Principle Behind the Derivation

- Determine the total flux linkage of one conductor (say A).

- The total flux linkage consists of:
  - Self flux linkage due to conductor A.
  - Mutual flux linkage due to conductor A' of the same phase.
  - Mutual flux linkages due to conductors B, B', C and C'.

- Using the principle of superposition, the individual flux linkages are added algebraically.

- Since the system is balanced,

$$I_A + I_B + I_C = 0$$

which simplifies the final expression.

Simplification

- The conductors of the same phase are connected electrically in parallel.

- Hence, the inductance obtained for one conductor is divided equally between the two conductors.

Inductance per phase = (Inductance per conductor)/2

- The final inductance depends only on:
  - Conductor GMR (r')
  - Spacing between adjacent conductors (d)

Thus, the six-conductor system is represented by an equivalent inductance expression for one phase.

Flux-linkages of phase A conductors

$$\psi_A = 2 \times 10^{-7}\left[I_A\left(\log_e \frac{1}{r'} + \log_e \frac{1}{2d}\right) + I_B\left(\log_e \frac{1}{d} + \log_e \frac{1}{\sqrt{3}d}\right) + I_C\left(\log_e \frac{1}{\sqrt{3}d} + \log_e \frac{1}{d}\right)\right]$$

$$= 2 \times 10^{-7}\left[I_A \log_e \frac{1}{2dr'} + (I_B + I_C)\log_e \frac{1}{\sqrt{3}d^2}\right]$$

$$= 2 \times 10^{-7}\left[I_A \log_e \frac{1}{2dr'} - I_A \log_e \frac{1}{\sqrt{3}d^2}\right]$$

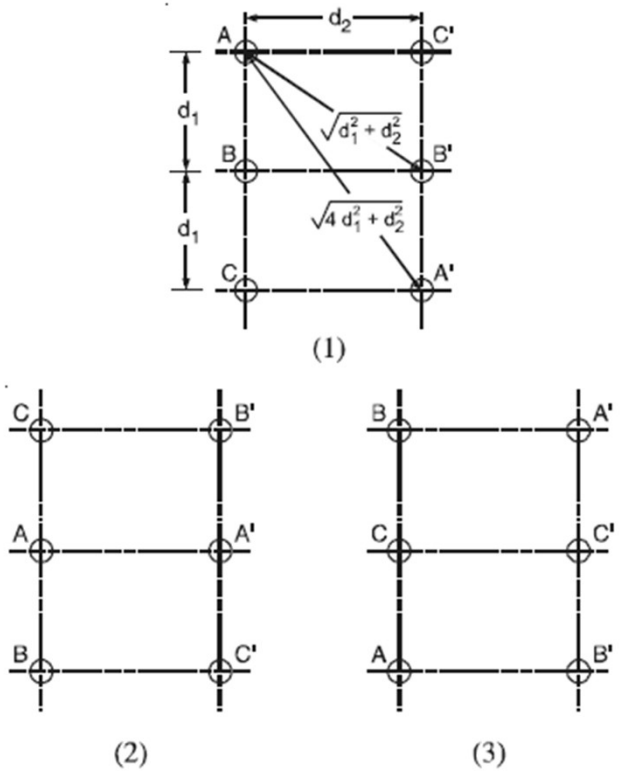
$$\because I_A + I_B + I_C = 0$$

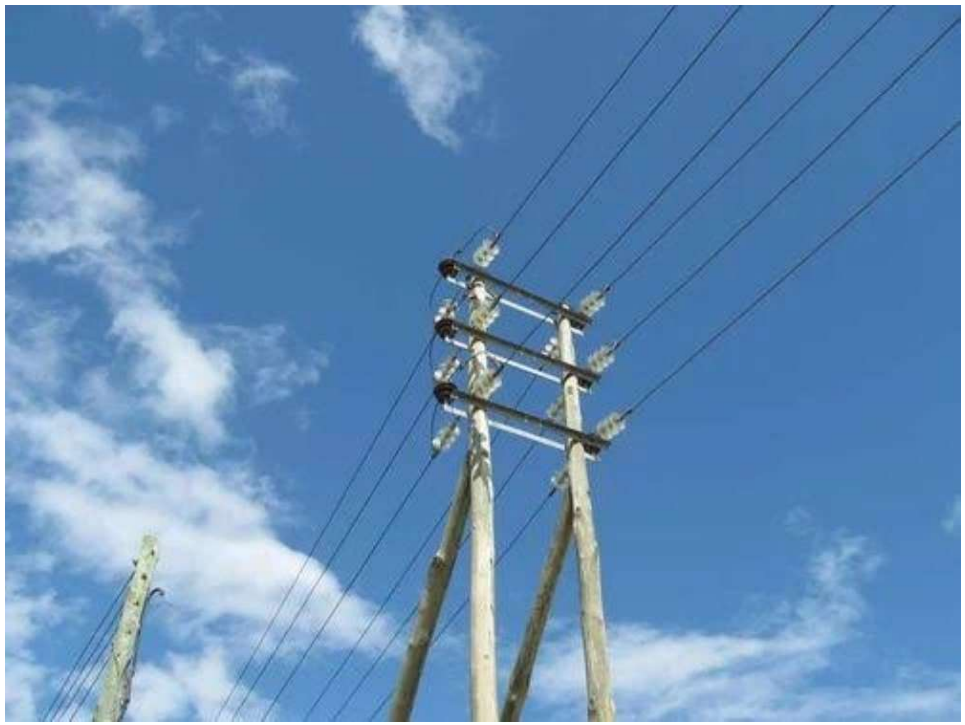
$$= 2 \times 10^{-7} I_A \log_e \frac{\sqrt{3}d}{2r'} \text{ Wb-turns/m}$$

Since conductors are electrically in parallel, inductance of each phase

$$= \frac{1}{2} L_A = 1 \times 10^{-7} \log_e \frac{\sqrt{3}d}{2r'} \text{ H/m} \qquad \ldots(5.30)$$

Inductance of a 3-Phase Double Circuit with Unsymmetrical Spacing but Transposed





Principle Behind the Derivation

- In a double-circuit line, each phase consists of two conductors connected in parallel.

- Due to unsymmetrical spacing, the magnetic coupling between conductors varies with their physical positions.

- During transposition, each phase occupies all three positions for one-third of the line length.

- The flux linkage of conductor A is determined separately for Positions 1, 2 and 3 by considering:
  - Self flux linkage.
  - Mutual flux linkage due to the parallel conductor (A').
  - Mutual flux linkages due to the remaining four phase conductors.

- The average flux linkage of conductor A is obtained as

$$\psi_A = (\psi_{A1} + \psi_{A2} + \psi_{A3})/3$$

Simplification of the Derivation

- The three flux-linkage expressions differ only in the spacing terms.

- Averaging the expressions removes the effect of conductor position and combines the spacing terms into a single equivalent expression.

- The inductance per conductor is obtained from

$$L_A = \psi_A / I_A$$

- Since two conductors form one phase and are electrically in parallel,

$$\text{Inductance per phase} = L_A / 2$$

- The final inductance depends only on:
  - Conductor GMR (r')
  - Vertical spacing (d1)
  - Horizontal spacing (d2)

Equivalent GMD and GMR Interpretation

- The final inductance expression can be written in the form
  $L = 2 \times 10^{-7} \ln$ (Equivalent GMD / Equivalent GMR)

- Comparing with the derived expression,
  $L = 2 \times 10^{-7} \ln [2^{1/6} (d_1/r')^{1/2}((d_1^2+d_2^2)/(4d_1^2+d_2^2))^{1/6}]$

- Equivalent Mutual GMD
  $D_{eq} = 2^{1/6} d_1^{1/2}(d_1^2 + d_2^2)^{1/6}$

- Equivalent Self GMR
  $D_{seq} = r'^{1/2}(4d_1^2 + d_2^2)^{1/6}$

- Hence,
  $L = 2 \times 10^{-7} \ln (Deq / Dseq)$ H/m

- If d_2 is large compared to d_1, $L = 2 \times 10^{-7} \log_e 2^{1/6} \left(\frac{d_1}{r'}\right)^{1/2}$ H/m

Interpretation of the Final Derivation

• The derivation shows that the inductance of a transposed 3-phase double-circuit line can also be expressed in the familiar form

$$L = 2 \times 10^{-7} \ln (Dm / Ds)$$

• Here, Dm and Ds are the equivalent Mutual GMD and Self GMD of the double-circuit line.

• The equivalent Self GMD is obtained by combining the Self GMDs of the three phase groups:

$$Ds = \sqrt[3]{(Dsa\ Dsb\ Dsc)}$$

• The equivalent Mutual GMD is obtained by combining the Mutual GMDs between every pair of phase groups:

$$Dm = \sqrt[3]{(Dab\ Dbc\ Dca)}$$

• Thus, irrespective of the conductor configuration, the inductance is always determined by the ratio of the equivalent Mutual GMD to the equivalent Self GMD.

Procedure for Evaluating Inductance of a Transposed 3-Phase Double-Circuit Line

- Determine the Self GMD (D_{sa}, D_{sb}, and D_{sc}) of the three phase groups.
- Determine the Mutual GMD (D_{ab}, D_{bc} and D_{ca}) between every pair of phase groups.
- Calculate the equivalent Self GMD and Mutual GMD of the transposed line.

  $D_s = \sqrt[3]{(D_{sa}\, D_{sb}\, D_{sc})}$

  $D_m = \sqrt[3]{(D_{ab}\, D_{bc}\, D_{ca})}$

- Obtain the inductance per phase using

  $L = 2 \times 10^{-7} \ln(D_m/D_s)$

$$\psi_{A1} = 2 \times 10^{-7} \left[I_A \left(\log_e \frac{1}{r'} + \log_e \frac{1}{\sqrt{4\,d_1^2 + d_2^2}} \right) + I_B \left(\log_e \frac{1}{d_1} + \log_e \frac{1}{\sqrt{d_1^2 + d_2^2}} \right) + I_C \left(\log_e \frac{1}{2d_1} + \log_e \frac{1}{d_2} \right) \right]$$

Similarly flux linkages with conductor A in position (2) and (3)

$$\psi_{A2} = 2 \times 10^{-7} \left[I_A \left(\log_e \frac{1}{r'} + \log_e \frac{1}{d_2} \right) + I_B \left(\log_e \frac{1}{d_1} + \log_e \frac{1}{\sqrt{d_1^2 + d_2^2}} \right) + I_C \left(\log_e \frac{1}{d_1} + \log_e \frac{1}{\sqrt{d_1^2 + d_2^2}} \right) \right]$$

$$\psi_{A3} = 2 \times 10^{-7} \left[I_A \left(\log_e \frac{1}{r'} + \log_e \frac{1}{\sqrt{4\,d_1^2 + d_2^2}} \right) + I_B \left(\log_e \frac{1}{2d_1} + \log_e \frac{1}{d_2} \right) + I_C \left(\log_e \frac{1}{d_1} + \log_e \frac{1}{\sqrt{d_1^2 + d_2^2}} \right) \right]$$

Average flux linkages with conductor A

$$\psi_A = \frac{\Psi_{A1} + \Psi_{A2} + \Psi_{A3}}{3}$$

$$\text{or } \psi_A = \frac{2 \times 10^{-7}}{3} I_A \log_e \frac{2\,d_1^3 \left(d_1^2 + d_2^2\right) d_2}{(r')^3\, d_2 \left(4\,d_1^2 + d_2^2\right)}$$

$$\text{or } \psi_A = 2 \times 10^{-7}\, I_A \log_e \frac{2^{1/3}\, d_1 \left(d_1^2 + d_2^2\right)^{1/3}}{r' \left(4\,d_1^2 + d_2^2\right)^{1/3}} \text{ wb-turns per metre}$$

and inductance,

$$L_A = \frac{\Psi_A}{I_A} = 2 \times 10^{-7} \log_e \frac{2^{1/3}\, d_1 \left(d_1^2 + d_2^2\right)^{1/3}}{r' \left(4\,d_1^2 + d_2^2\right)^{1/3}} \text{ H/m}$$

$$= 2 \times 10^{-7} \log_e 2^{1/3} \frac{d_1}{r'} \left(\frac{d_1^2 + d_2^2}{4\,d_1^2 + d_2^2} \right)^{1/3} \text{ H/m} \quad \ldots(5.31)$$

Inductance of each phase,

$$L = \frac{1}{2} \times \text{inductance per conductor}$$

$$= \frac{1}{2} L_A$$

$$= 2 \times 10^{-7} \log_e 2^{1/6} \left(\frac{d_1}{r'}\right)^{1/2} \left(\frac{d_1^2 + d_2^2}{4 d_1^2 + d_2^2}\right)^{1/6} \text{ H/m} \quad ...(5.32)$$

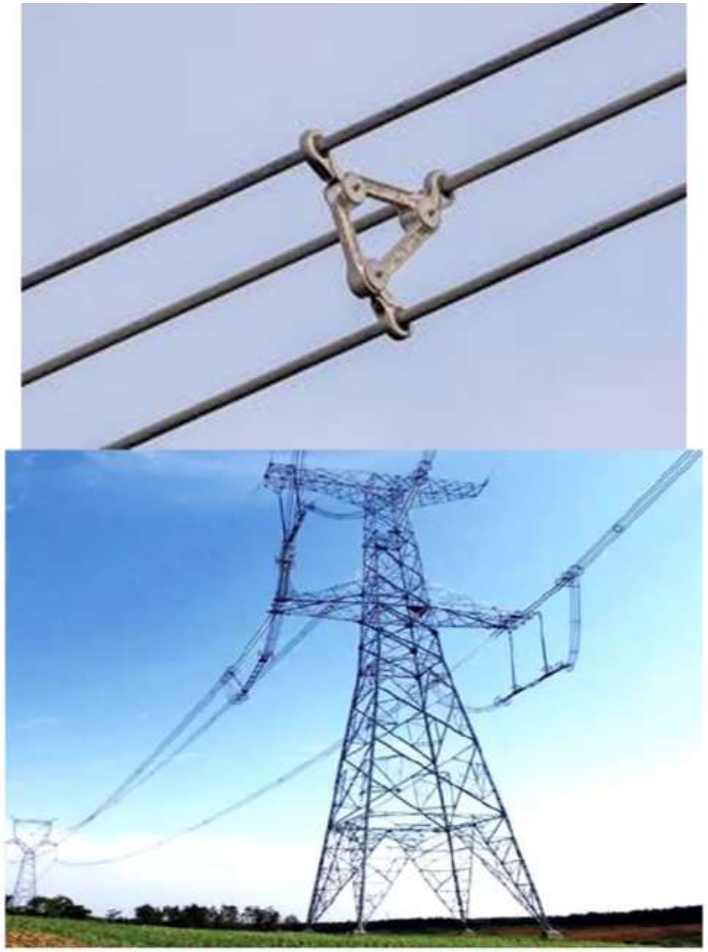
If the distance d_2 is too large as compared to d_1, $\frac{d_1^2 + d_2^2}{4d_1^2 + d_2^2}$ would tend to be unity and inductance per phase,

$$L = 2 \times 10^{-7} \log_e 2^{1/6} \left(\frac{d_1}{r'}\right)^{1/2} \text{ H/m}$$

$$\text{or } L = 0.2 \log_e 2^{1/6} \left(\frac{d_1}{r'}\right)^{1/2} \text{ mH/km} \quad ...(5.33)$$

Bundled Conductor

- Two or more sub-conductors are grouped to act as a single phase in high-voltage transmission systems.
- Spaced at uniform intervals by metallic spacers
- they increase effective surface area, significantly reducing power loss, line inductance, and dangerous corona discharge



Single Circuit Transmission Line with Bundled Conductors

Bundled Conductors

- A bundled conductor consists of two or more sub-conductors connected electrically in parallel to form one phase conductor.
- Bundle conductors are used in EHV and UHV transmission lines (typically 220 kV and above).

- Reduce corona loss and radio interference.
- Increase the effective GMR (Self GMD) of the phase.
- Reduce the inductance and inductive reactance of the line.
- Increase the line capacitance and surge impedance loading (SIL).
- Improve voltage regulation and increase power transfer capability.

Typical Configurations

- Duplex bundle – 2 sub-conductors
- Triplex bundle – 3 sub-conductors
- Quadruplex bundle – 4 sub-conductors

Principle Behind the Derivation

- Each sub-conductor carries an equal share of the phase current.

- Every sub-conductor is linked by:
  - Flux due to its own current.
  - Flux due to currents in the other sub-conductors.

- Using the principle of superposition, the total flux linkage of each sub-conductor is obtained.

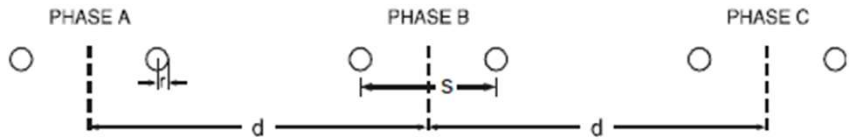
- The equivalent GMR (Self GMD) of the bundle is then calculated and used to determine the inductance of the bundled conductor.

GMR, GMD and Inductance of Bundled Conductors

For sub-conductor GMR = r′ and bundle spacing = s:

Duplex: $D_s = \sqrt{(r's)}$

Triplex: $D_s = \sqrt[3]{(r's^2)}$



Quadruplex: $D_s = 1.09\sqrt[4]{(r's^3)}$

Dm = geometric mean of the distances between the phase bundles.

Inductance per phase:

$L = 2 \times 10^{-7} \ln(D_m/D_s)$ H/m

Higher Ds → Lower inductance and reactance

Composite Conductors vs. Bundled Conductors

Composite Conductors

- Consist of several strands forming one conductor.
- Used for mechanical strength and flexibility.
- Equivalent GMR depends on strand geometry.

Bundled Conductors

- Consist of two or more complete conductors forming one phase.
- Used in EHV/UHV transmission lines.
- Equivalent GMR depends on bundle spacing.

In both cases, the final inductance expression is

$$L = 2 \times 10^{-7} \ln(D_m / D_s) \text{ H/m}$$

Only the method of calculating D_s is different.

Example 5.1. Find the loop inductance and reactance per km of a single-phase overhead line consisting of two conductors, each 1.25 cm diameter. The spacing between conductors is 1 m and frequency is 50 Hz.

Solution : Conductor of radius, $r = \frac{1.25}{2}$ cm = 0.625 cm

Spacing of conductors, d = 1 m = 100 cm

Geometric mean radius (GMR) of conductor,

$r' = 0.7788\ r = 0.7788 \times 0.625 = 0.48675$ cm

Loop inductance per km of a single phase line

$$L = 0.4 \log_e \frac{d}{r'} \text{ mH} \qquad \text{Refer to Eq. (5.9)}$$

$$= 0.4 \log_e \frac{100}{0.48675} = 2.13 \text{ mH } \textbf{Ans.}$$

Reactance of transmission line,

$$X_L = 2\pi fL = 2\pi \times 50 \times 2.13 \times 10^{-3} = 0.669\ \Omega/\text{km } \textbf{Ans.}$$

Example 5.2. Calculate inductance of a 500 m long, 1-ϕ, 2-wire transmission line. Each wire has identical area of cross section of 10 cm^2 and conductors are separated by a distance of 5 m.

Solution : *Conductor radius,* $r = \sqrt{\frac{a}{\pi}} = \sqrt{\frac{10}{\pi}} = 1.784$ cm $\because a = \pi r^2$

Spacing of conductors, $d = 5$ m

Geometric mean radius (GMR) of conductor,

$$r' = 0.7788r = 0.7788 \times 1.784 = 1.39 \text{ cm}$$

Loop inductance of 500 m long single phase line,

$$L = 0.4 \log_e \frac{d}{r'} \times l = 0.4 \log_e \frac{500}{1.39} \times \frac{1}{2} = 1.177 \text{ mH}$$ **Ans.**

Example 5.3. **Calculate the loop inductance per km of a single-phase line comprising of 2 parallel conductors 1 metre apart and 1 cm in diameter, when the material of conductor is (*i*) copper and (*ii*) steel of relative permeability 50.**

Solution : Conductor radius, $r = 0.5$ cm

Spacing of conductors, $d = 1$ metre $= 100$ cm

(*i*) Loop inductance per km of line with copper conductor

$$= \left(1 + 4\log_e \frac{d}{r}\right) \times 10^{-7} \times 1{,}000 \text{ H}$$

$$= \left(0.1 + 0.4\log_e \frac{100}{0.5}\right) \times 10^{-3} \text{ H} = 2.22 \text{ mH}$$ **Ans.**

(*ii*) Loop inductance per km of line with steel conductor

$$= \left(\mu + 4\log_e \frac{d}{r}\right) \times 10^{-7} \times 1{,}000 \text{ H}$$

$$= \left(50 + 4\log_e \frac{100}{0.5}\right) \times 10^{-4} \text{ H} = 7.12 \text{ mH}$$ **Ans.**

Example 5.4. A single-phase transmission line has two parallel conductors, each of 1.2 cm diameter and 2.5 m apart. Calculate the loop inductance per km length of line if the material of the conductor is steel with relative permeability of 200.

Solution : Loop inductance per km of line with steel conductor

$$= \left(\mu + 4 \log_e \frac{d}{r}\right) \times 10^{-7} \times 1{,}000 \text{ H}$$

$$= \left(200 + 4 \log_e \frac{250}{0.6}\right) \times 10^{-4} \text{ H} = 22.4 \text{ mH}$$ **Ans.**

Example 5.5. Calculate the GMR of a 6/3 mm Al, 1/3 mm steel ACSR conductor.[*A.M.I.E. Sec. B. Power Systems Winter 1996*]

Solution : Figure 5.8 shows an ACSR conductor with 6 outer conductors of aluminium each of radius $r = 3$ mm and one central conductor of steel also of radius $r = 3$ mm.

From Fig. 5.8

$d_{11} = r$; $d_{12} = d_{16} = d_{17} = 2r$; $d_{14} = 4r$ and

$$d_{13} = d_{15} = \sqrt{D_{14}^2 - D_{34}^2} = \sqrt{(4r)^2 - (2r)^2} = 2\sqrt{3}\, r$$

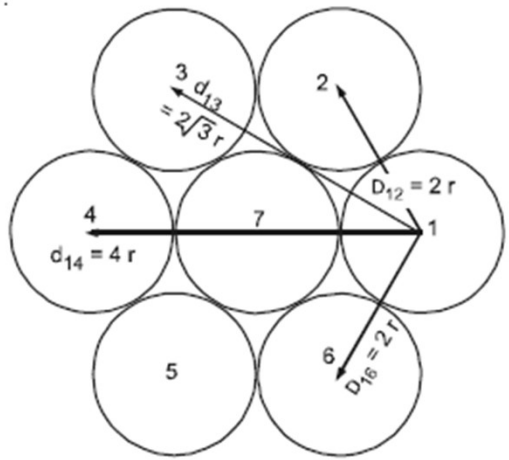


Fig. 5.8. ***Cross-Section of a 7-Strand Conductor***

$$D_{s1} = D_{s2} = D_{s3} = D_{s4} = D_{s5} = D_{s6}$$

$$= \sqrt[7]{0.7788r \times 2r \times 2\sqrt{3}\,r \times 4r \times 2\sqrt{3}\,r \times 2r \times 2r} = \sqrt[7]{299}\,r$$

$$D_{s7} = \sqrt[7]{2r \times 0.7788r \times 2r \times 2r \times 2r \times 2r \times 2r} = \sqrt[7]{49.8432}\,r$$

Geometric mean radius,

$$D_s = \sqrt[7]{D_{s1} \cdot D_{s2} \cdot D_{s3} \cdot D_{s4} \cdot D_{s5} \cdot D_{s6} \cdot D_{s7}}$$

$$= r\sqrt[49]{299^6 \times 49.8432}$$

$$= 2.176r = 2.176 \times 3 \text{ mm} = 6.528 \text{ mm}$$ **Ans.**

Example 5.6. A split-phase, single phase transmission line is shown in Fig. 5.9. Conductors 1 and 2 in parallel form one path while conductors 1′ and 2′ in parallel form the return path. The current is equally shared by the two parallel conductors. Determine the total inductance per km of the line. The radius of each conductor is 1.2 cm.

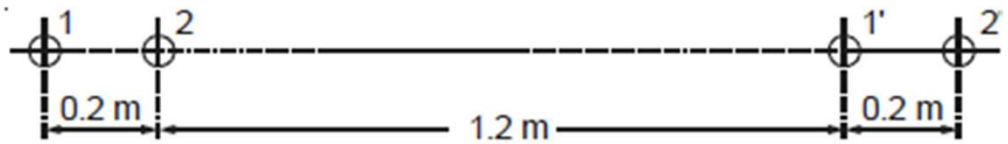


Fig. 5.9

Solution : Let the current flowing in each conductor be I amperes

Radius of each conductor, $r = 1.2$ cm

Geometric mean radius, $r' = 0.7788 \times 1.2 = 0.93456$ cm

Spacings of conductors, $d_{12} = 0.2$ m $= 20$ cm

$d_{11'} = (0.2 + 1.2)$ m $= 140$ cm ; $d_{22'} = (0.2 + 1.2)$ m $= 140$ cm

$d_{12'} = (0.2 + 1.2 + 0.2)$ m $= 160$ cm ; $d_{21'} = 1.2$ m $= 120$ cm

Mutual GMD,

$$D_m = \sqrt[4]{d_{11'}\, d_{12'}\, d_{21'}\, d_{22'}} = \sqrt[4]{140 \times 160 \times 120 \times 140}$$

$$= 139.28 \text{ cm}$$

Self GMD,

$$D_s = \sqrt[4]{d_{11}\, d_{12}\, d_{21}\, d_{22}} = \sqrt[4]{0.93456 \times 20 \times 20 \times 0.93456}$$

$$= 4.32 \text{ cm}$$

Loop inductance of line,

$$L = 0.4 \log_e \frac{D_m}{D_s} \text{ mH/km}$$

$$= 0.4 \log_e \frac{139.28}{4.32} = 1.389 \text{ mH/km}$$ **Ans.**

Example 5.8. Calculate the inductance per km of a 3-phase overhead transmission line using 1.24 cm diameter conductors when these are placed at the corners of an equilateral triangle of each side 2 metres.

Solution : Conductor radius, $r = \frac{1.24}{2}$ cm $= 0.62$ cm

Geometric mean radius $r' = 0.7788r = 0.7788 \times 0.62$ cm

$= 0.4829$ cm

Spacing of conductors, $d = 2$ m $= 200$ cm

Inductance per phase per km $= 0.2 \log_e \frac{d}{r'}$ mH

$= 0.2 \log_e \frac{200}{0.4829}$ mH

$= 1.205$ mH **Ans.**

Example 5.9. Determine the inductance per km of a 3-ϕ transmission line using 20 mm diameter conductors when conductors are situated at the corners of a triangle with spacing of 4, 5 and 6 metres. Conductors are regularly transposed.

Solution : Conductor radius, $r = \frac{20}{2} = 10 \text{ mm} = 1.0 \text{ cm}$

Geometric mean radius, $r' = 0.7788 \times 1.0 = 0.7788 \text{ cm}$

Inductance per phase,

$$L = 0.2 \log_e \frac{(d_1\, d_2\, d_3)^{1/3}}{r'} \text{ mH/km}$$

$$= 0.2 \log_e \frac{(400 \times 500 \times 600)^{1/3}}{0.7788} \text{ mH} = 1.29 \text{ mH}$$ **Ans.**

Example 5.10. The three conductors of a 3-phase transmission line are arranged in a horizontal plane and are 3 metres apart. The diameter of each conductor is 4 cm. Determine the inductance per km of each phase. Assume balanced load and R, Y, B phase sequence.

Solution : Radius of each conductor, $r = \frac{4}{2}$ cm = 2 cm

GMR, $r' = 0.7788 \times 2$ cm = 1.5576 cm

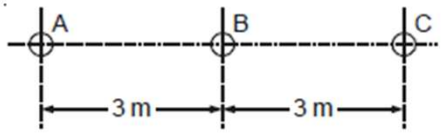


Fig. 5.16

From Eqs. (5.23), (5.24) and (5.25), respectively

$$L_A = 0.2\left[\log_e \frac{d}{r'} + \frac{1}{2}\log_e 2 - j0.866 \log_e 2\right] \text{ mH/km}$$

$$= 0.2\left[\log_e \frac{300}{1.5576} + \frac{1}{2}\log_e 2 - j\,0.866 \log_e 2\right] \text{ mH/km}$$

$$= 0.2\ [5.26 + 0.346 - j0.866 \times 0.6931]$$

$$= (1.1212 - j0.12) \text{ mH}$$ **Ans.**

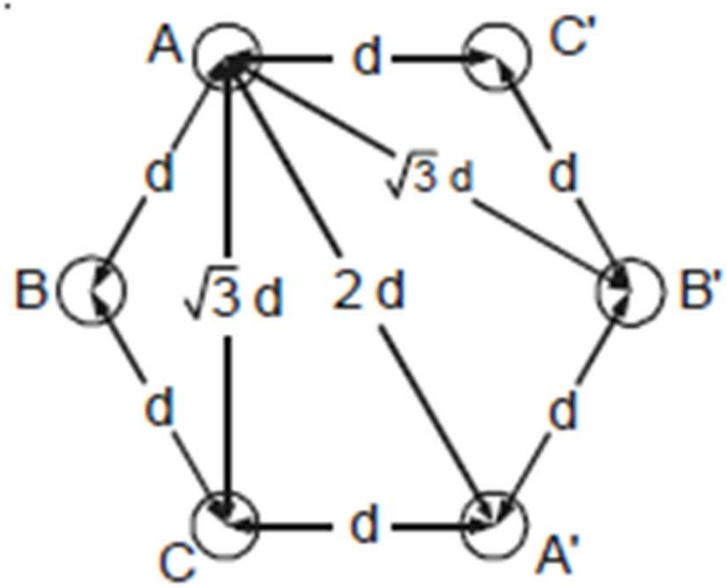
$$L_B = 0.2 \log_e \frac{d}{r'} \text{ mH/km} = 0.2 \log_e \frac{300}{1.5576} = 1.052 \text{ mH}$$ **Ans.**

$$L_C = 0.2\left[\log_e \frac{d}{r'} + \frac{1}{2}\log_e 2 + j0.866 \log_e 2\right] \text{ mH}$$

$$= 0.2\left[\log_e \frac{300}{1.5576} + \frac{1}{2}\log_e 2 + j0.866 \log_e 2\right] \text{ mH}$$

$$= (1.1212 + j0.12) \text{ mH}$$ **Ans.**

Example 5.12. **Determine the inductance per phase per km of a double circuit 3-phase line. The radius of each conductor is 20 mm and the conductors are placed on the circumference of an imaginary circle of radius 7 m forming a regular hexagonal figure.**[*A.M.I.E.*



Solution : Refer to Fig. 5.19.

d = 7 m or 7,000 mm

GMR, r' = 0.7788 r = 0.7788 × 20 mm = 15.576 mm

From Eq. (5.30), inductance per phase

$$L = 1 \times 10^{-7} \log_e \frac{\sqrt{3}}{2} \frac{d}{r'} \text{ H/m}$$

$$= 0.1 \log_e \frac{\sqrt{3}}{2} \frac{d}{r'} \text{ mH/km}$$

$$= 0.1 \log_e 0.866 \times \frac{7{,}000}{15.576} = 0.5964 \text{ mH}$$ **Ans.**

Example 5.13. A part of the transposition cycle of a 3-phase double circuit line is shown in Fig. 5.21. Radius of each conductor is 0.9 cm. The conductors are solid copper. Find the inductance per phase per km of line. [*A.M.I.E. Sec. B. Power Systems Winter 1996*]

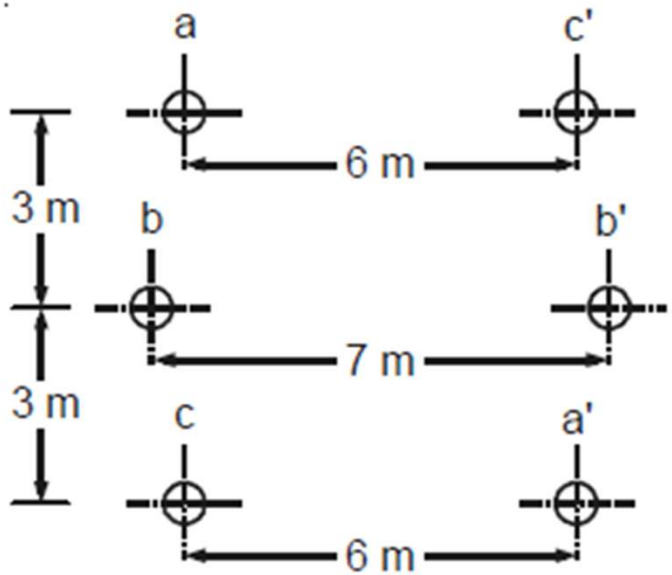
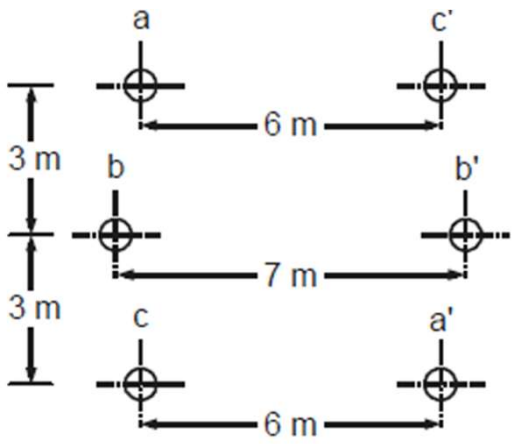


Fig. 5.21



Solution : GMR of each conductor

$r' = 0.7788 \times 0.9 \text{ cm} = 0.7 \text{ cm} = 0.7 \times 10^{-2} \text{ m}$

Distance between conductors a and a',

$$D_{aa'} = \sqrt{6^2 + 6^2} = 8.485 \text{ m}$$

$d_{bb'} = 7$ m and $d_{cc'} = d_{aa'} = 8.485$ m

GMR of conductors of phase a,

$$D_{sa} = 4\sqrt{d_{aa}\, d_{aa'}\, d_{a'a'}\, d_{a'a}}$$

$$= \sqrt{0.7\times10^{-2} \times 8.485 \times 0.7 \times 10^{-2} \times 8.485}$$

$$= 0.2437 \text{ m}$$

Similarly,

$$D_{sb} = \sqrt{0.7 \times 10^{-2} \times 7} = 0.2214 \text{ m}$$

and $D_{sc} = \sqrt{0.7 \times 10^{-2} \times 8.485} = 0.2437$ m

Self GMD,

$$D_s = \sqrt[3]{D_{sa}\, D_{sb}\, D_{sc}} = \sqrt[3]{0.2437 \times 0.2214 \times 0.2437} = 0.236 \text{ m}$$

Distance between conductors a and b,

$$d_{ab} = \sqrt{3^2 + 0.5^2} = 3.041 \text{ m}$$

Distance between conductors a and b',

$$d_{ab'} = \sqrt{3^2 + 6.5^2} = 7.159 \text{ m}$$

$$D_{ab} = \sqrt[4]{d_{ab}\, d_{ab'}\, d_{a'b}\, d_{a'b'}}$$

$$= \sqrt[4]{3.041 \times 7.159 \times 7.159 \times 3.041} = 4.666 \text{ m}$$

Similarly,

$$D_{bc} = \sqrt[4]{(3.041 \times 7.159)^2} = 4.666 \text{ m and}$$

$$D_{ca} = \sqrt[4]{(6 \times 6)^2} = 6 \text{ m}$$

$$D_{eq} \text{ or } D_m = \sqrt[3]{D_{ab}\, D_{bc}\, D_{ca}} = \sqrt[3]{4.666 \times 4.666 \times 6} = 5.074 \text{ m}$$

Inductance per phase,

$$L = 0.2 \log_e \frac{D_m}{D_s} \text{ mH/km} = 0.2 \times \log_e \frac{5.074}{0.236}$$

$$= 0.6136 \text{ mH/km}$$ **Ans.**

Example 5.18. Determine the inductance per km per phase of a single circuit 460 kV line using two bundle conductors per phase as shown in the figure. The diameter of each conductor is 5.0 cm.

[U.P. Technical Univ. Elements of Power System 2007-08]

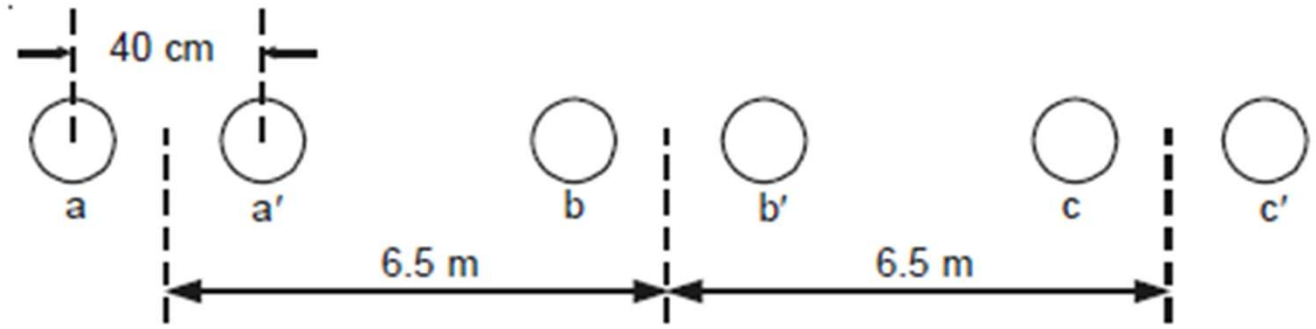
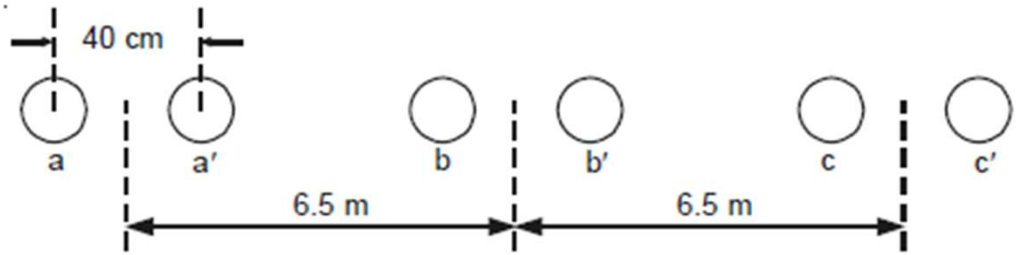


Fig. 5.28



Solution : Radius of each sub-conductor, $r = \frac{5.0}{2}$ cm = 2.5 cm

GMR of each sub-conductor,

$$r' = 0.7788r = 0.7788 \times 2.5 \text{ cm} = 1.947 \text{ cm} = 1.947 \times 10^{-2} \text{ m}$$

Phase-to-phase separation,

$$d = 6.5 \text{ m}$$

Spacing between sub-conductors of one phase,

$$s = 0.4 \text{ m}$$

$$\text{GMR, } D_s = \sqrt{r's} = \sqrt{1.947 \times 10^{-2} \times 0.4} = 0.088 \text{ m}$$

$$\text{GMR, } D_s = \sqrt{r's} = \sqrt{1.947 \times 10^{-2} \times 0.4} = 0.088 \text{ m}$$

$$D_{ab} = D_{bc} = \sqrt[4]{d_{ab}d_{ab'}d_{a'b}d_{a'b'}} = \sqrt[4]{6.5 \times 6.9 \times 6.1 \times 6.5}$$

$$= 6.494 \text{ m}$$

$$D_{ca} = \sqrt[4]{d_{ca}d_{ca'}d_{c'a}d_{c'a'}} = \sqrt[4]{13 \times 12.6 \times 13.4 \times 13} = 12.997 \text{ m}$$

$$D_{eq} \text{ or } D_m = \sqrt[3]{D_{ab}D_{bc}D_{ca}} = \sqrt[3]{6.494 \times 6.494 \times 12.997} = 8.184 \text{ m}$$

Taking the centre to centre distances between the bundles as D_{ab}, D_{bc} and D_{ca}

$$D_{eq} = \sqrt[3]{6.5 \times 6.5 \times 13} = 8.189 \text{ m}$$

which is almost the same, as the accurate value of D_{eq}.

Inductance of the bundled conductor line per phase,

$$L = 0.2 \log_e \frac{D_m}{D_s} \text{ mH/km} = 0.2 \log_e \frac{8.184}{0.088}$$

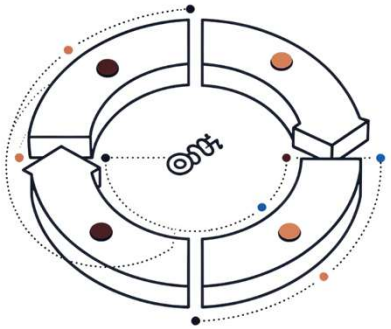
$$= 0.907 \text{ mH/km}$$ **Ans.**

Transposition of Double Circuit Lines

What Is Transposition?

Transposition **cyclically rotates conductor positions** along the line to equalize impedances and eliminate sustained unbalanced voltages due to asymmetric spacing.

- Each conductor occupies every position for exactly **one-third** of the total line length
- Three sections of equal length: **L/3 each**
- Conductors a, b, c, a′, b′, c′ rotate through all six positions across the three sections



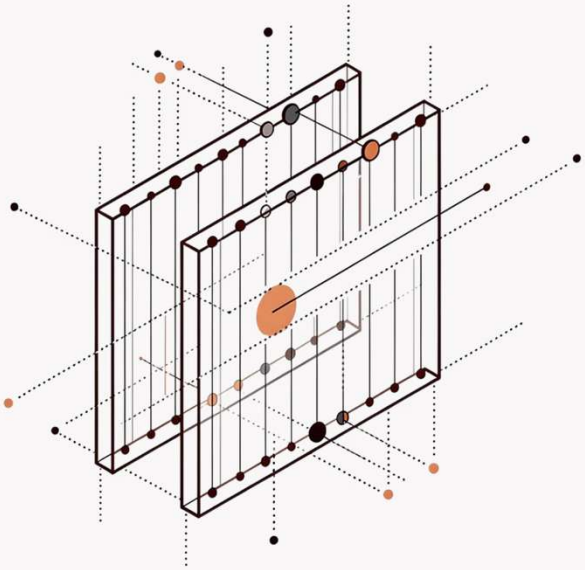
Why It Matters for Derivation

Fully transposed lines permit the use of **Geometric Mean Distance (GMD)** and **Geometric Mean Radius (GMR)** in closed-form parameter formulas — the central simplification used throughout this derivation.

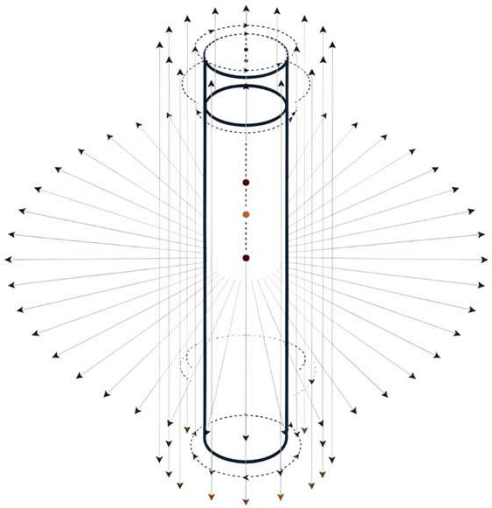
CHAPTER 3

Capacitance

Electric field fundamentals, line-to-neutral capacitance derivation, double circuit extension, and earth correction via method of images.



Electric Field and Potential — Fundamentals



Electric Field of a Long Charged Conductor

A conductor with linear charge density q (C/m) produces a **radial electric field** at distance r:

> $\mathbf{E = q / (2\pi\varepsilon r)}$ (V/m)

where $\varepsilon = \varepsilon_0 = 8.854 \times 10^{-12}$ F/m for air. The field is directed radially outward for positive charge.

Potential Difference Between Two Points

Integrating the electric field from r_1 to r_2:

> $\mathbf{V12 = (q / 2\pi\varepsilon) \cdot \ln(r_2 / r_1)}$ (V)

Superposition for Three-Phase Lines

The voltage at any conductor is the algebraic sum of contributions from all three phase charges qa, qb, qc. For a balanced system: qa + qb + qc = 0, which simplifies the expression significantly.

Transmission Line Capacitance - Significance

- Transmission line capacitance stores electric energy in the electric field between conductors and between conductors and earth.

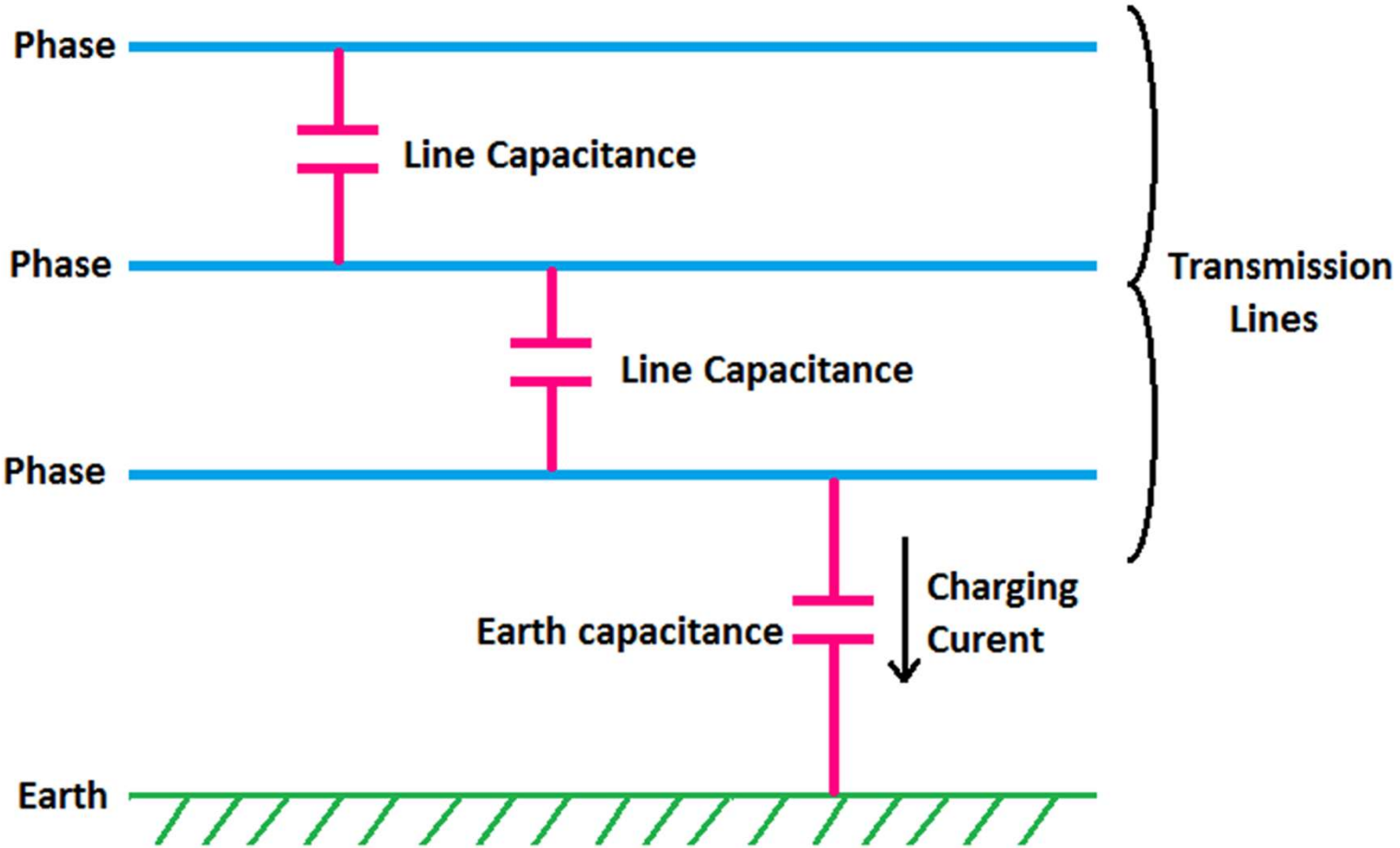
- It draws a charging current even when the receiving end is open-circuited.

- The charging current supplies reactive power to the system, influencing the voltage profile.

- Line capacitance is responsible for the Ferranti Effect, where the receiving-end voltage exceeds the sending-end voltage under light-load conditions.

- It affects the surge impedance and Surge Impedance Loading (SIL) of the transmission line.

- Capacitance increases the power transfer capability of EHV transmission lines and therefore becomes an important design parameter.


Phase
Line Capacitance
Phase
Line Capacitance
Phase
Transmission Lines
Charging Curent
Earth capacitance
Earth

Capacitance — Three-Phase Single Circuit

1 Apply Superposition

For a balanced three-phase system with qa + qb + qc = 0, the voltage of phase a with respect to neutral is derived by summing potential contributions from all three charged conductors.

2 Phase Voltage Expression

After applying transposition (replacing unequal spacings with Deq) and using the balanced charge condition:

$$V_{an} = \frac{q_a}{2\pi\varepsilon} \cdot \ln\left(\frac{D_{eq}}{r}\right)$$

3 Line-to-Neutral Capacitance

Dividing charge by voltage gives the capacitance per unit length:

$$C_n = \frac{2\pi\varepsilon}{\ln(D_{eq} / r)} \ \text{(F/m)}$$

Charging current per phase: $I_c = j\omega C_n V_{an}$ — significant for long EHV lines where it can reach tens of amperes per kilometre.

Capacitance — Three-Phase Double Circuit Derivation

Setup: Six Conductors

Label conductors 1–6 with charge densities qa, qb, qc, qa′, qb′, qc′. By symmetry of the double circuit: qa = qa′, qb = qb′, qc = qc′. Balanced condition: qa + qb + qc = 0.

Voltage of Phase a After Transposition

Superposing all six conductor contributions and averaging over the three transposition sections:

> **Van = (qa / 2πε) · ln(Deq / Ds)**

Here Ds and Deq are the same equivalent GMR and GMD as derived for inductance.

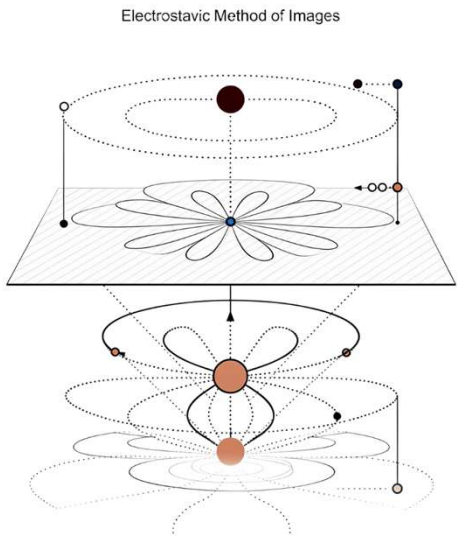
Final Capacitance Formula

> **Cn = 2πε / ln(Deq / Ds)** (F/m)

Since the double circuit effectively doubles the charge per unit voltage, the equivalent capacitance per phase is:

> **Ceq = 2Cn** — double that of a single circuit.

Effect of Earth — Method of Images



The Problem: Earth as a Conductor

The earth surface is an equipotential plane. A conductor at height h above ground induces surface charges that modify the electric field distribution — and therefore the capacitance.

Method of Images

The earth plane is replaced by **image conductors** located at mirror-image positions below the ground surface, carrying **opposite charges** (−qa, −qb, −qc). This preserves the equipotential boundary condition at ground level.

Effect on Capacitance

- Image conductors modify the potential equations — the **equivalent GMD increases slightly**
- A larger Deq means a larger denominator in ln(Deq/Ds), slightly **reducing capacitance**
- For high-voltage lines, earth effect on capacitance is typically **~1–3%** — small but non-negligible in precise studies

Gauss's Law

The total electric flux through a closed surface equals the enclosed charge divided by the permittivity of the medium:

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{enclosed}}}{\varepsilon_0 \varepsilon_r}$$

Consider a long cylindrical conductor of radius r carrying a uniform charge q C/m.

Substitute the components back into Gauss's Law:

$$E \cdot (2\pi x L) = \frac{q \cdot L}{\varepsilon_0 \varepsilon_r}$$

The length (L) cancels out from both sides, yielding the final formula:

$$E = \frac{q}{2\pi \varepsilon_0 \varepsilon_r x}$$

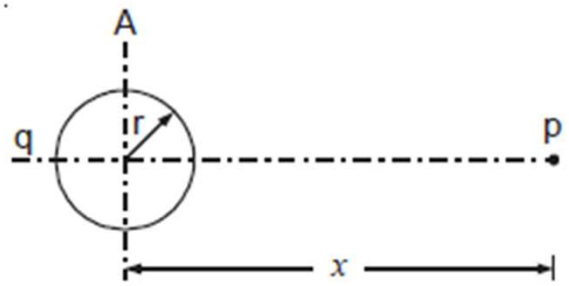
Potential of a Charged Single Conductor

- Consider a long cylindrical conductor of radius r carrying a uniform charge q C/m.

- The potential of the conductor is obtained by integrating the electric field from the conductor surface to infinity.

$V_A = \int_r^\infty E\, dx$

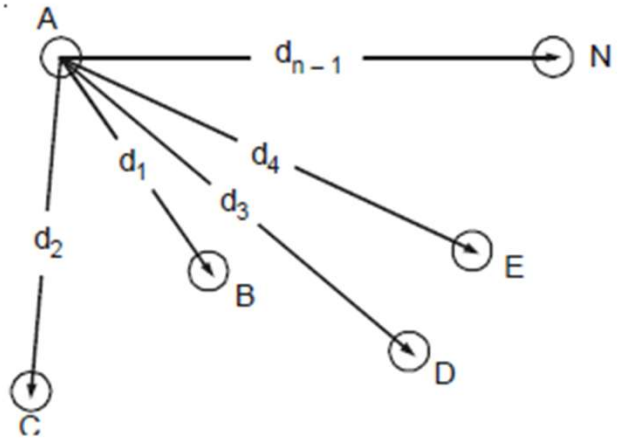
$V_A = (q / 2\pi \varepsilon_0) \int_r^\infty (dx/x)$



$$V_A = \int_r^\infty \frac{q}{2\pi \in_0 x} dx = \frac{q}{2\pi \in_0} \int_r^\infty \frac{dx}{x}$$

Potential at a Charged Conductor in a Group of Charged Conductors

Consider a group of long straight conductors A, B, C, D, E, ..., N having charges of q_1, q_2, q_3, q_4, q_5, ..., q_n coulombs per metre length respectively (Fig. 5.33).



$$V_{AN} = \frac{1}{2\pi \in_0}\left[q_1 \log_e \frac{1}{r} + q_2 \log_e \frac{1}{d_1} + q_3 \log_e \frac{1}{d_2} + \ldots + q_n \log_e \frac{1}{d_{n-1}}\right] \text{ volts} \quad \ldots(5.39)$$

Potential of conductor A due to its own charge q_1

$$= \int_{r}^{\infty} \frac{q_1}{2\pi \epsilon_0} \frac{dx}{x} \text{ volts}$$

Potential of conductor A due to charge q_2

$$= \int_{d_1}^{\infty} \frac{q_2}{2\pi \epsilon_0} \frac{dx}{x} \text{ volts}$$

since the field due to the charge q_2 extends from infinity up to a distance d_1 from conductor A.

Similarly, potential of conductor A, due to charge q_3 of conductor C placed at a distance of d_2 metres from conductor A

$$= \int_{d_2}^{\infty} \frac{q_3}{2\pi \epsilon_0} \frac{dx}{x} \text{ volts}$$

Potential of conductor A due to charge q_n of conductor N placed at a distance of d_{n-1} metres from conductor A

$$= \int_{d_{n-1}}^{\infty} \frac{q_n}{2\pi \epsilon_0} \frac{dx}{x} \text{ volts}$$

So overall potential difference between conductor A and infinite distant neutral plane

$$V_{AN} = \int_{r}^{\infty} \frac{q_1}{2\pi \epsilon_0} \frac{dx}{x} + \int_{d_1}^{\infty} \frac{q_2}{2\pi \epsilon_0} \frac{dx}{x} + \int_{d_2}^{\infty} \frac{q_3}{2\pi \epsilon_0} \frac{dx}{x} + \cdots + \int_{d_{n-1}}^{\infty} \frac{q_n}{2\pi \epsilon_0} \frac{dx}{x}$$

$$= \frac{1}{2\pi \epsilon_0} \left[q_1 (\log_e \infty - \log_e r) + q_2 (\log_e \infty - \log_e d_1) + q_3 (\log_e \infty - \log_e d_3) + \ldots + q_n (\log_e \infty - \log_e d_{n-1}) \right]$$

$$= \frac{1}{2\pi \epsilon_0} \left[q_1 \log_e \frac{1}{r} + q_2 \log_e \frac{1}{d_1} + q_3 \log_e \frac{1}{d_2} + \ldots. + q_n \log_e \frac{1}{d_{n-1}} + \log_e \infty (q_1 + q_2 + q_3 + \ldots. + q_n) \right]$$

Assuming balanced load *i.e.*, $q_1 + q_2 + q_3 \ldots q_n = 0$

or

$$V_{AN} = \frac{1}{2\pi \epsilon_0} \left[q_1 \log_e \frac{1}{r} + q_2 \log_e \frac{1}{d_1} + q_3 \log_e \frac{1}{d_2} + \ldots + q_n \log_e \frac{1}{d_{n-1}} \right] \text{ volts} \quad \ldots(5.39)$$

Capacitance of a Single-Phase Transmission Line

• The capacitance of a transmission line arises due to the electric field between the conductors.

• The electric field stores electrostatic energy and gives rise to line capacitance.

Capacitance between line and earth

$C_{L\text{-}E} = 2\pi\varepsilon / \ln(d/r)$ F/m

$C_{L\text{-}L} = \dfrac{\pi \in_0}{\log_e \dfrac{d}{r}}$ F/m

Observation

Larger conductor spacing decreases capacitance, whereas larger conductor radius increases capacitance.

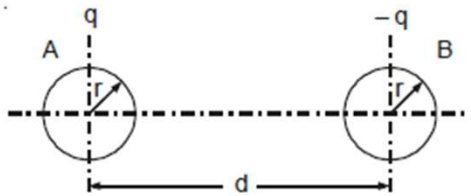


Fig. 5.34

PD between conductor A and neutral 'infinite' plane,

$$V_{AN} = \int_r^\infty \frac{q}{2\pi \in_0} \cdot \frac{d}{dx} + \int_d^\infty \frac{-q}{2\pi \in_0} \frac{dx}{x}$$

$$= \frac{q}{2\pi \in_0} \left[\log_e \infty + \log_e \frac{1}{r} - \log_e \infty + \log_e d \right]$$

$$= \frac{q}{2\pi \in_0} \log_e \frac{d}{r}$$

Similarly *pd* between conductor B and neutral 'infinite' plane,

$$V_{BN} = \int_r^\infty \frac{-q}{2\pi \in_0} \frac{dx}{x} + \int_d^\infty \frac{q}{2\pi \in_0} \frac{dx}{x} = \frac{-q}{2\pi \in_0} \log_e \frac{d}{r} \text{ volts}$$

PD between conductors A and B,

$$V_{AB} = V_{AN} - V_{BN}$$

$$= \frac{q}{2\pi \in_0} \log_e \frac{d}{r} - \frac{-q}{2\pi \in_0} \log_e \frac{d}{r} = \frac{q}{\pi \in_0} \log_e \frac{d}{r} \text{ volts}$$

Capacitance of the line,

$$C = \frac{q}{V_{AB}} = \frac{q}{\frac{q}{\pi \in_0} \log_e \frac{d}{r}} = \frac{\pi \in_0}{\log_e \frac{d}{r}} \text{ F/m} \qquad \ldots(5.40)$$

CAPACITANCE OF 3-PHASE OVERHEAD LINES

• In a 3-phase line, each conductor carries an equal and opposite charge with respect to the neutral.

• The potential of a conductor is the algebraic sum of the potentials produced by the charges on all three conductors.

• Since the conductor spacing is unequal (d1, d2 and d3), the capacitance of each phase is different.

• Therefore, the capacitances of phases A, B and C are calculated separately.

• The capacitance of each phase is obtained from

C = q / V

where q is the charge per metre and V is the phase-to-neutral potential.

$$C_{AN} = 2\pi\epsilon_0 \frac{\log_e \frac{d_2}{r} - \frac{V_{BN}}{V_{AN}} \log_e \frac{d_3}{d_1}}{\log_e \frac{d_3}{r} \log_e \frac{d_2}{r} - \log_e \frac{d_2}{d_1} \cdot \log_e \frac{d_3}{d_1}} \text{ F/m}$$

Principle Behind the Derivation

- Determine the phase-to-neutral potential of each conductor using the principle of superposition.

- The potential of a conductor consists of:
  - Potential due to its own charge.
  - Potential due to charges on the other two conductors.

- Since the system is balanced,

$$q_A + q_B + q_C = 0$$

this relation is used to simplify the potential equations.

- The capacitance of each phase is then obtained using

$$C = q / V$$

- Because the conductor spacing is unsymmetrical, the capacitances of the three phases are generally unequal.

$$C_{AN} = \frac{q_1}{V_{AN}}$$

$$= 2\pi\epsilon_0 \frac{\log_e \frac{d_2}{r} - \frac{V_{BN}}{V_{AN}} \log_e \frac{d_3}{d_1}}{\log_e \frac{d_3}{r} \log_e \frac{d_2}{r} - \log_e \frac{d_2}{d_1} \cdot \log_e \frac{d_3}{d_1}} \text{ F/m} \quad ...(5.49)$$

Similarly capacitance of conductor B to neutral,

$$C_{BN} = 2\pi\epsilon_0 \frac{\log_e \frac{d_3}{r} - \frac{V_{CN}}{V_{BN}} \log_e \frac{d_1}{d_2}}{\log_e \frac{d_1}{r} \cdot \log_e \frac{d_3}{r} - \log_e \frac{d_3}{d_2} \cdot \log_e \frac{d_1}{d_2}} \text{ F/m} \quad ...(5.50)$$

and

$$C_{CN} = 2\pi\epsilon_0 \frac{\log_e \frac{d_1}{r} - \frac{V_{AN}}{V_{CN}} \log_e \frac{d_2}{d_3}}{\log_e \frac{d_2}{r} \cdot \log_e \frac{d_1}{r} - \log_e \frac{d_1}{d_3} \cdot \log_e \frac{d_2}{d_3}} \text{ F/m} \quad ...(5.51)$$

Capacitance of Equilaterally Spaced 3 Phase Lines

For the equilateral spacing d1 = d2 = d3 = d (say)

$$C_{AN} = C_{BN} = C_{CN} = C_N = \frac{2\pi \epsilon_0}{\log_e \frac{\sqrt[3]{d.d.d}}{r}} = \frac{2\pi \epsilon_0}{\log_e \frac{d}{r}}$$

Capacitance of Unsymmetrical Line With Transposed Conductors

Assuming that the charge per unit length is same in every part of the transposed cycle [Figs. 5.35 (1), (2) and (3)] average value of voltage of conductor A, will be

$$\frac{V_{AN_1} + V_{AN_2} + V_{AN_3}}{3} \qquad \text{...(5.52)}$$

Voltage of conductor A in positions (1), (2), and (3) we have

$$V_{AN_1} = \frac{1}{2\pi \epsilon_0}\left[q_1 \log_e \frac{1}{r} + q_2 \log_e \frac{1}{d_1} + q_3 \log_e \frac{1}{d_3}\right] \text{ volts}$$

$$V_{AN_2} = \frac{1}{2\pi \epsilon_0}\left[q_1 \log_e \frac{1}{r} + q_2 \log_e \frac{1}{d_2} + q_3 \log_e \frac{1}{d_1}\right] \text{ volts}$$

and

$$V_{AN_3} = \frac{1}{2\pi \epsilon_0}\left[q_1 \log_e \frac{1}{r} + q_2 \log_e \frac{1}{d_3} + q_3 \log_e \frac{1}{d_2}\right] \text{ volts}$$

$$C_{AN} = C_{BN} = C_{CN} = C_N = \frac{2\pi \epsilon_0}{\log_e \dfrac{\sqrt[3]{d_1 d_2 d_3}}{r}} \text{ F/m}$$

So average value of voltage of conductor A,

$$V_{AN} = \frac{V_{AN_1} + V_{AN_2} + V_{AN_3}}{3}$$

$$= \frac{1}{6\pi \epsilon_0}\left[q_1 \log_e \frac{1}{r^3} + q_2 \log_e \frac{1}{d_1 d_2 d_3} + q_3 \log_e \frac{1}{d_1 d_2 d_3}\right]$$

$$= \frac{1}{6\pi \epsilon_0}\left[q_1 \log_e \frac{1}{r^3} + (q_2 + q_3) \log_e \frac{1}{d_1\ d_2\ d_3}\right] \text{ volts}$$

Substituting $q_2 + q_3 = -q_1$ in above equation, we have

$$V_{AN} = \frac{1}{6\pi \epsilon_0}\left[q_1 \log_e \frac{1}{r^3} - \log_e \frac{1}{d_1\ d_2\ d_3}\right] \text{ volts}$$

$$\text{or } V_{AN} = \frac{q_1}{2\pi \epsilon_0}\left[\log_e \frac{1}{r} + \log_e \sqrt[3]{d_1\ d_2\ d_3}\right]$$

$$= \frac{q_1}{2\pi \epsilon_0} \log_e \frac{\sqrt[3]{d_1\ d_2\ d_3}}{r} \text{ volts}$$

Capacitance of conductor A to neutral,

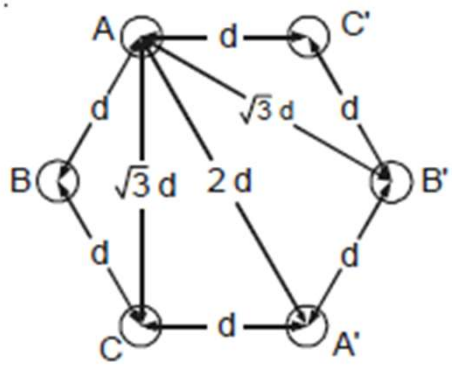
$$C_{AN} = \frac{q_1}{V_{AN}} = \frac{q_1}{\frac{q_1}{2\pi \epsilon_0} \log_e \frac{\sqrt[3]{d_1 d_2 d_3}}{r}} = \frac{2\pi \epsilon_0}{\log_e \frac{\sqrt[3]{d_1 d_2 d_3}}{r}} \text{ F/m}$$

Similarly expressions for C_{BN} and C_{CN} can be obtained and we have

$$C_{AN} = C_{BN} = C_{CN} = C_N = \frac{2\pi \epsilon_0}{\log_e \frac{\sqrt[3]{d_1 d_2 d_3}}{r}} \text{ F/m} \qquad ...(5.53)$$

CAPACITANCE OF DOUBLE CIRCUIT 3-PHASE OVERHEAD LINES

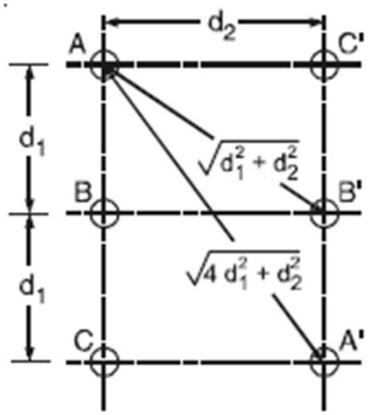
Symmetrically Spaced Line



$$C_{AN} = C_{BN} = C_{CN} = C_N = \frac{2\pi\epsilon_0}{\log_e \frac{\sqrt{3}d}{2r}} \text{ F/m}$$

$$\text{Capacitance per phase, } C = 2C_N = \frac{4\pi\epsilon_0}{\log_e \frac{\sqrt{3}d}{2r}} \text{ F/m}$$

Flat Vertically Spaced Line



Symmetrically Spaced Line

Potential of conductor A *w.r.t.* neutral infinite plane (Fig. 5.19),

$$V_{AN} = \frac{1}{2\pi\epsilon_0}\left[q_1\int_r^\infty \frac{dx}{x} + q_2\int_d^\infty \frac{dx}{x} + q_3\int_{\sqrt{3}\,d}^\infty \frac{dx}{x} + q_1\int_{2d}^\infty \frac{dx}{x} + q_2\int_{\sqrt{3}\,d}^\infty \frac{dx}{x} + q_3\int_d^\infty \frac{dx}{x}\right]$$

$$= \frac{1}{2\pi\epsilon_0}\left[q_1\log_e\infty - q_1\log_e r + q_2\log_e\infty - q_2\log_e d - q_3\log_e\sqrt{3}d + q_3\log_e\infty + q_1\log_e\infty - q_1\log_e 2d + q_2\log_e\infty - q_2\log_e\sqrt{3}\,d + q_3\log_e\infty - q_3\log_e d\right]$$

$$= \frac{1}{2\pi\epsilon_0}\left[2(q_1+q_2+q_3)\log_e\infty + q_1\log_e\frac{1}{2\,dr} + q_2\log_e\frac{1}{\sqrt{3}\,d^2} + q_3\log_e\frac{1}{\sqrt{3}\,d^2}\right]$$

$$= \frac{1}{2\pi\epsilon_0}\left[q_1\log_e\frac{1}{2\,dr} + q_2\log_e\frac{1}{\sqrt{3}\,d^2} + q_3\log_e\frac{1}{\sqrt{3}\,d^2}\right]$$

$\because\ q_1 + q_2 + q_3 = 0$

$$= \frac{1}{2\pi\epsilon_0}\left[q_1\log_e\frac{1}{2\,dr} + (q_2+q_3)\log_e\frac{1}{\sqrt{3}\,d^2}\right]$$

$$= \frac{1}{2\pi\epsilon_0}\left[q_1\log_e\frac{1}{2\,dr} - q_1\log_e\frac{1}{\sqrt{3}\,d^2}\right] \qquad \because\ q_2 + q_3 = -q_1$$

$$= \frac{q_1}{2\pi\epsilon_0}\log_e\frac{\sqrt{3}d^2}{2\,dr} = \frac{q_1}{2\pi\epsilon_0}\log_e\frac{\sqrt{3}d}{2r} \text{ volts}$$

Capacitance of conductor A to neutral,

$$C_{AN} = \frac{q_1}{V_{AN}} = \frac{q_1}{\frac{q_1}{2\pi\epsilon_0}\log_e\frac{\sqrt{3}\,d}{2r}} = \frac{2\pi\epsilon_0}{\log_e\frac{\sqrt{3}\,d}{2r}} \text{ F/m}$$

$$C_{AN} = C_{BN} = C_{CN} = C_N = \frac{2\pi\epsilon_0}{\log_e\frac{\sqrt{3}d}{2r}} \text{ F/m}$$

Flat Vertically Spaced Line

$$V_{AN_1} = \frac{1}{2\pi \epsilon_0}\left[q_1 \int_r^\infty \frac{dx}{x} + q_2 \int_{d_1}^\infty \frac{dx}{x} + q_3 \int_{2d_1}^\infty \frac{dx}{x} + q_1 \int_{\sqrt{4d_1^2 + d_2^2}}^\infty \frac{dx}{x} + q_2 \int_{\sqrt{d_1^2 + d_2^2}}^\infty \frac{dx}{x} + q_3 \int_{d_2}^\infty \frac{dx}{x} \right] \text{volts}$$

$$= \frac{1}{2\pi \epsilon_0}\left[q_1 \log_e \frac{1}{r\sqrt{4d_1^2 + d_2^2}} + q_2 \log_e \frac{1}{d_1\sqrt{d_1^2 + d_2^2}} + q_3 \log_e \frac{1}{2d_1 d_2} \right] \text{volts}$$

Similarly,

$$V_{AN_2} = \frac{1}{2\pi \epsilon_0}\left[q_1 \log_e \frac{1}{r d_2} + q_2 \log_e \frac{1}{d_1\sqrt{d_1^2 + d_2^2}} + q_3 \log_e \frac{1}{d_1\sqrt{d_1^2 + d_2^2}} \right] \text{volts and}$$

$$V_{AN_3} = \frac{1}{2\pi \epsilon_0}\left[q_1 \log_e \frac{1}{r\sqrt{4d_1^2 + d_2^2}} + q_2 \log_e \frac{1}{2d_1 d_2} + q_3 \log_e \frac{1}{d_1\sqrt{d_1^2 + d_2^2}} \right]$$

$$V_{AN} = \frac{V_{AN1} + V_{AN2} + V_{AN3}}{3}$$

$$= \frac{1}{6\pi \epsilon_0}\left[q_1 \log_e \frac{1}{r^3 d_2 (4d_1^2 + d_2^2)} + q_2 \log_e \frac{1}{2d_1^3 d_2 (d_1^2 + d_2^2)} + q_3 \log_e \frac{1}{2d_1^3 d_2 (d_1^2 + d_2^2)} \right]$$

$$= \frac{1}{6\pi \epsilon_0}\left[q_1 \log_e \frac{1}{r^3 d_2 (4d_1^2 + d_2^2)} + (q_2 + q_3) \log_e \frac{1}{2d_1^3 d_2 (d_1^2 + d_2^2)} \right]$$

$$= \frac{1}{6\pi \epsilon_0}\left[q_1 \log_e \frac{1}{r^3 d_2 (4d_1^2 + d_2^2)} - q_1 \log_e \frac{1}{2d_1^3 d_2 (d_1^2 + d_2^2)} \right]$$

$\because\ q_2 + q_3 = -\ q_1$

$$= \frac{q_1}{6\pi\epsilon_0} \log_e \frac{2d_1^3 d_2 (d_1^2 + d_2^2)}{r^3 d_2 (4d_1^2 + d_2^2)}$$

$$= \frac{q_1}{6\pi\epsilon_0} \log_e \frac{2\,d_1^3}{r^3} \cdot \frac{d_1^2 + d_2^2}{4d_1^2 + d_2^2}$$

$$= \frac{q_1}{2\pi\epsilon_0} \log_e \frac{2^{1/3} d_1}{r} \left(\frac{d_1^2 + d_2^2}{4d_1^2 + d_2^2} \right)^{1/3} \text{ volts}$$

Capacitance of conductor A,

$$C_{AN} = \frac{q_1}{V_{AN}} = \frac{q}{\frac{q}{2\pi\epsilon_0} \log_e 2^{1/3} \frac{d_1}{r} \left(\frac{d_1^2 + d_2^2}{4d_1^2 + d_2^2} \right)^{1/3}}$$

$$= \frac{2\pi\epsilon_0}{\log_e 2^{1/3} \frac{d_1}{r} \left(\frac{d_1^2 + d_2^2}{4d_1^2 + d^2} \right)^{1/3}} \text{ F/m}$$

Capacitance per phase will be double of CAN, i.e.,

Capacitance per phase

$$C = \frac{4\pi\epsilon_0}{\log_e 2^{1/3} \frac{d_1}{r} \left(\frac{d_1^2 + d_2^2}{4d_1^2 + d_2^2} \right)^{1/3}} \text{ F/m}$$

CAPACITANCE OF BUNDLED CONDUCTOR LINE

The conductors of any one bundle are in parallel and it is assumed that the charge per bundle divides equally among the conductors of the bundle as Dab >> s. Also Dab – s j Dab + s j Dab for the same reason.

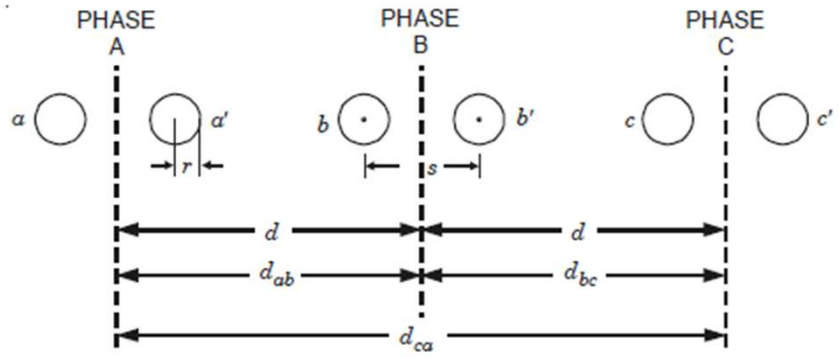


Fig. 5.40. *Cross Section of a Bundled Conductor Three-Phase Transmission Line*

$$C_N = \frac{2\pi\varepsilon_0}{\log_e \frac{\sqrt[3]{d_{ab}d_{bc}d_{ca}}}{\sqrt{rs}}} = \frac{2\pi\varepsilon_0}{\log_e \frac{D_m}{D_s}}$$

$$D_m = \sqrt[3]{d_{ab}d_{bc}d_{ca}}$$

Example 5.20. **Find out capacitance of a single phase line 30 km long consisting of two parallel wires each 15 mm diameter and 1.5 m apart.**

Solution : Radius of each conductor, $r = \frac{15}{2}$ mm = 7.5 mm

Spacing between conductors, d = 1.5 m = 1,500 mm

Capacitance of line,

$$C = \frac{\pi \epsilon_0}{\log_e \frac{d}{r}} \times \text{length of line in metres}$$

$$= \frac{\pi \times 8.854 \times 10^{-12}}{\log_e \frac{1{,}500}{7.5}} \times 30{,}000 = 0.1575\ \mu\text{F}$$ **Ans.**

Example 5.21. Calculate the capacitance of a 100 km long 3-phase, 50 Hz overhead transmission line consisting of 3-conductors, each of diameter 2 cm and spaced 2.5 m at the corners of equilateral triangle.

Solution : Spacing of conductors, $d = 2.5$ m $= 250$ c

Radius of each conductor, $r = \frac{2.0}{2} = 1.0$ cm

$$\text{Capacitance of the line} = \frac{2\pi \epsilon_0}{\log_e \frac{d}{r}} \text{ F/m}$$

$$= \frac{2\pi \times 8.854 \times 10^{-12}}{\log_e \frac{250}{1}}$$

$$= 10.075 \times 10^{-12} \text{ F/m}$$

Capacitance of 100 km long line

$$= 10.075 \times 100 \times 10^3 \times 10^{-12}$$

$$= 1.0075 \ \mu\text{F}$$ **Ans.**

Example 5.22. **A single 3-phase line operated at 50 Hz is arranged as follows : (Fig. 5.36). The conductor diameter is 0.6 cm. Find the inductance and capacitance per km. The line is regularly transposed.** [*Rajasthan Technical Univ. Transmission & Distribution of Electrical Power 2011*]

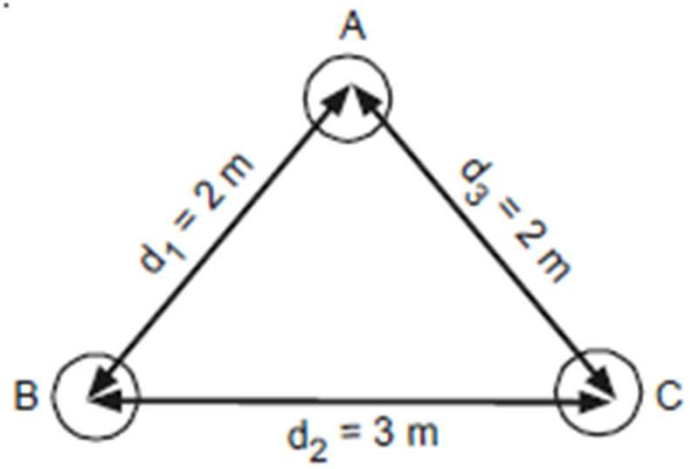


Fig. 5.36

Solution : Conductor radius, $r = \frac{0.6}{2}$ cm = 0.3 cm

Geometric mean radius, $r' = 0.7788 \times 0.3$ cm = 0.23364 cm

Spacing $d_1 = d_3 = 2.0$ m = 200 cm

$d_2 = 3$ m = 300 cm

Inductance per km per phase

$$= 0.2 \log_e \frac{\sqrt[3]{d_1 d_2 d_3}}{r'} \text{ mH}$$

$$= 0.2 \log_e \frac{\sqrt{200 \times 300 \times 200}}{0.23364} \text{ mH} = 1.378 \text{ mH}$$ **Ans.**

Capacitance per phase per metre

$$= \frac{2\pi \epsilon_0}{\log_e \frac{\sqrt[3]{d_1 d_2 d_3}}{r}} = \frac{2\pi \times 8.854 \times 10^{-12}}{\log_e \frac{\sqrt[3]{200 \times 300 \times 200}}{0.3}}$$

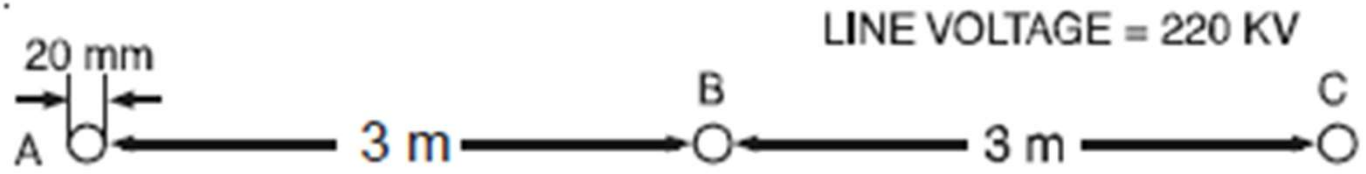
$$= 8.3814 \times 10^{-12} \text{ F}$$

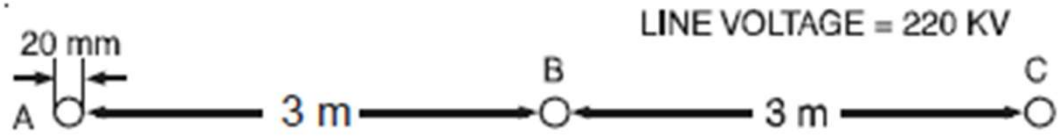
Capacitance per phase per km

$= 8.3814 \times 10^{-12} \times 1{,}000 = 8.3814 \times 10^{-9}$ F **Ans.**

Example 5.23. Determine the capacitance and charging current per unit length of the line for the following arrangement of conductors of diameter 20 mm shown in Fig. 5.37.

transposed





Solution : $r = \frac{20}{2} = 10 \text{ mm} = 0.01 \text{ m}$

Spacing between conductors

$$d_1 = \text{AB} = 3 \text{ m}; \; d_2 = \text{BC} = 3 \text{ m}; \; d_3 = \text{CA} = 6 \text{ m}$$

Capacitance per phase per m length,

$$C_N = \frac{2\pi \epsilon_0}{\log_e \frac{\sqrt[3]{d_1 d_2 d_3}}{r}} = \frac{2\pi \times 8.854 \times 10^{-12}}{\log_e \frac{\sqrt[3]{3 \times 3 \times 6}}{0.01}}$$

$$= 9.3737 \times 10^{-12} \text{ F } \textbf{Ans.}$$

Charging current per phase,

$$I_C = 2\pi f C_N V_{ph}$$

$$= 2\pi \times 50 \times 9.3737 \times 10^{-12} \times \frac{220 \times 1{,}000}{\sqrt{3}} = 0.374 \text{ mA } \textbf{Ans.}$$

Example 5.25. Six conductors of a double-circuit transmission line are arranged as shown below : (Fig. 5.38)

The diameter of each conductor is 2.5 cm. Find the capacitive reactance to neutral and the charging current per km per phase at 132 kV and 50 Hz, assuming that the line is regularly transposed. Neglect the effect of earth.

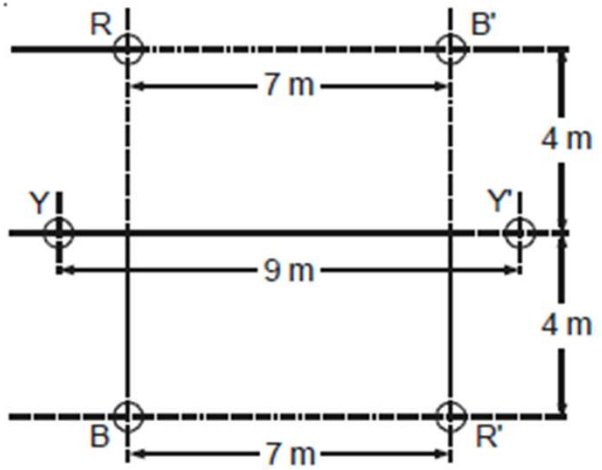


Fig. 5.38

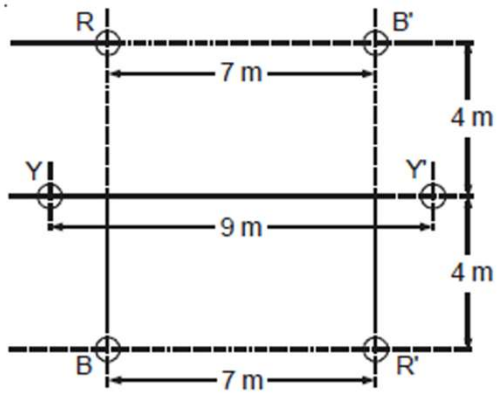


Fig. 5.38

$$d_{RR'} = \sqrt{7^2 + (4+4)^2} = 10.63 \text{ m}$$

$$d_{BB'} = d_{RR'} = 10.63 \text{ m}$$

GMR of conductors of phase R,

$$D_{SR} = \sqrt[4]{d_{RR}\, d_{RR'}\, d_{R'R'}\, d_{R'R}}$$

$$= \sqrt{r \times d_{RR'}} = \sqrt{1.25 \times 10^{-2} \times 10.63} = 0.3645 \text{ m}$$

Similarly, $D_{SY} = \sqrt{r \times d_{YY'}} = \sqrt{1.25 \times 10^{-2} \times 9} = 0.3354$ m

and $D_{SB} = \sqrt{r \times d_{BB'}} = \sqrt{1.25 \times 10^{-2} \times 10.63} = 0.3645$ m

Self, GMD, $D_S = \sqrt[3]{D_{SR}\, D_{SY}\, D_{SB}}$

$$= \sqrt[3]{0.3645 \times 0.3354 \times 0.3645} = 0.3545 \text{ m}$$

Distance between conductors R and Y,

$$D_{RY} = \sqrt{4^2 + (4.5 - 3.5)^2} = 4.12 \text{ m}$$

Distance between conductors R and Y′,

$$D_{RY'} = \sqrt{(9-1)^2 + 4^2} = 8.95 \text{ m}$$

$$\text{So } D_{RY} = \sqrt[4]{d_{RY}\, d_{RY'}\, d_{R'Y}\, d_{R'Y'}}$$

$$= \sqrt[4]{4.12 \times 8.95 \times 8.95 \times 4.12} = 6.072 \text{ m}$$

$$\text{Similarly, } D_{YB} = \sqrt[4]{(4.12 \times 8.95)^2} = 6.072 \text{ m}$$

$$\text{and } D_{BR} = \sqrt[4]{(8 \times 7)^2} = 7.483 \text{ m}$$

$$D_{eq} \text{ or } D_m = \sqrt[3]{D_{RY} D_{YB} D_{BR}}$$

$$= \sqrt[3]{6.072 \times 6.072 \times 7.483} = 6.51 \text{ m}$$

Capacitance per phase,

$$C = \frac{2\pi \epsilon_0}{\log_e \frac{D_m}{D_s}}$$

$$= \frac{2\pi \times 8.854 \times 10^{-12}}{\log_e \frac{6.51}{0.3545}}$$

$$= 19.11 \times 10^{-12} \text{ F/m or } 19.11 \times 10^{-9} \text{ F/km}$$

Capacitance reactance to neutral

$$= \frac{1}{2\pi f C} = \frac{1}{2\pi \times 50 \times 19.11 \times 10^{-9}}$$

$$= 166.526 \text{ k}\Omega \textbf{ Ans.}$$

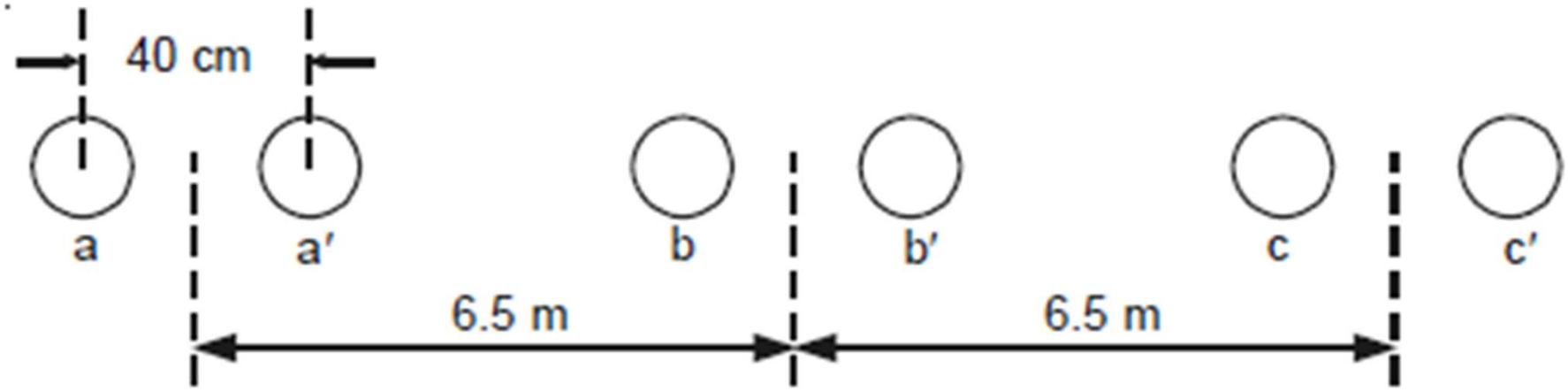
Charging current,

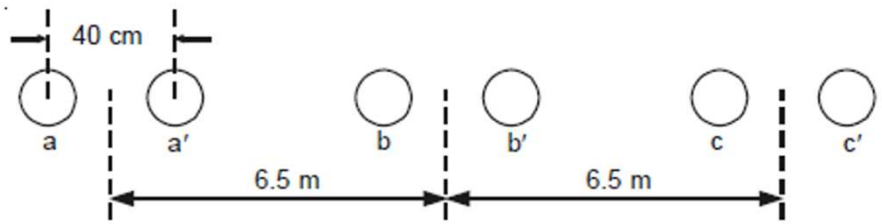
$$I_C = 2\pi f C V_{ph}$$

$$= 2\pi \times 50 \times 19.11 \times 10^{-9} \times \frac{132 \times 1{,}000}{\sqrt{3}}$$

$$= 0.4575 \text{ A/km } \textbf{Ans.}$$

Example 5.27. Determine the capacitance and charging current per km of the line of Example 5.18 if the line operates at 132 kV.





$$D_{ab} = D_{bc} = \sqrt[4]{d_{ab}d_{ab'}d_{a'b}d_{a'b'}} = \sqrt[4]{6.5 \times 6.9 \times 6.1 \times 6.5}$$

$$= 6.494 \text{ m}$$

$$D_{ca} = \sqrt[4]{d_{ca}d_{ca'}d_{c'a}d_{c'a'}} = \sqrt[4]{13 \times 12.6 \times 13.4 \times 13} = 12.997 \text{ m}$$

$$D_{eq} \text{ or } D_m = \sqrt[3]{D_{ab}D_{bc}D_{ca}} = \sqrt[3]{6.494 \times 6.494 \times 12.997} = 8.184 \text{ m}$$

Taking the centre to centre distances between the bundles as D_{ab}, D_{bc} and D_{ca}

$$D_{eq} = \sqrt[3]{6.5 \times 6.5 \times 13} = 8.189 \text{ m}$$

which is almost the same, as the accurate value of D_{eq}.

Solution : Self GMD, $D_s = \sqrt{rs} = \sqrt{2.5 \times 10^{-2} \times 0.4} = 0.1$ m

Mutual GMD, $D_m = 8.189$ m

Capacitance per km per phase, C

$$= \frac{2\pi\varepsilon_0}{\log_e \frac{D_m}{D_s}}$$

$$= \frac{2\pi \times 8.854 \times 10^{-12}}{\log_e \frac{8.189}{0.1}} \text{ F/m}$$

$$= 12.628 \times 10^{-12} \text{ F/m or } 12.628 \times 10^{-9} \text{ F/km}$$

Charging current per km

$$= 2\pi f C V_{ph} = 2\pi \times 50 \times 12.628 \times 10^{-9} \times \frac{132{,}000}{\sqrt{3}}$$

assuming supply frequency $f = 50$ Hz

$$= 0.3 \text{ A/km}$$ **Ans.**

Key Takeaways & Next Steps

GMD/GMR — The Core Tool

Double circuit lines require careful GMD and GMR computation accounting for **inter-circuit mutual coupling** across all six conductor positions.

Transposition Is Critical

Full transposition is the key assumption that enables **closed-form parameter expressions** — without it, the system remains unbalanced and unsymmetrical.

Feeds System Models

Derived R, L, C parameters feed directly into the **π-model** or **distributed parameter (Bergeron) model** for power flow and transient stability analysis.

Next: ABCD Constants

Apply these parameters to compute **ABCD constants** and analyze voltage regulation, efficiency, and surge impedance loading of the double circuit line.

